Chapter 4-Part 2: Cell Membranes

- What is a cell membrane made of & what does it do?
- What molecules can & can not cross cell membranes on their own?
- How do molecules know whether to move in or out of a cell?
- What are the possible ways to move across a cell membrane?

Corresponds with OpenStax Biology 2e Chapter 5

What is a cell membrane made of & what does it do?

- All living things are composed of cells & all cells have a cell membrane
  - <u>Cell membranes have 3 components:</u>
    - Phospholipids
    - Cholesterol
    - Proteins

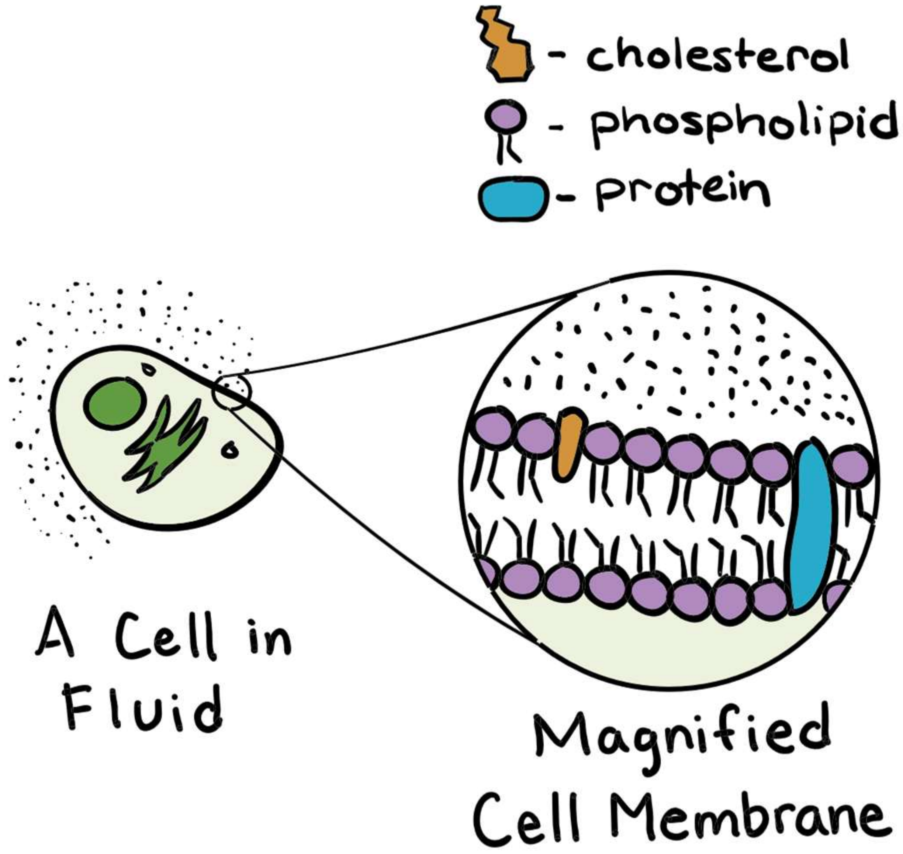
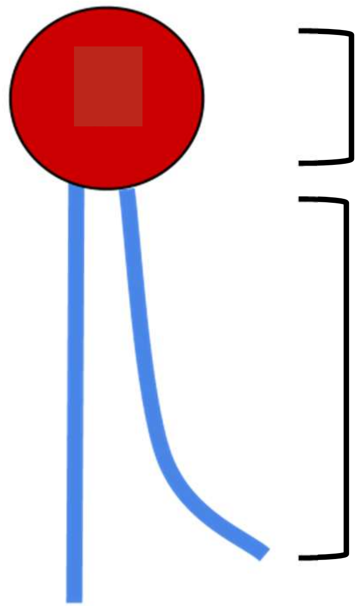


Figure: https://www.khanacademy.org/science/ap-biology/cell-structure-and-function/membrane-permeability/a/fluid-mosaic-model-cell-membranes-article

- Component 1: **Phospholipids** – a unique lipid
  - Remember: lipids are hydrophobic
    - However: **phospholipids** have hydrophobic AND hydrophilic portions



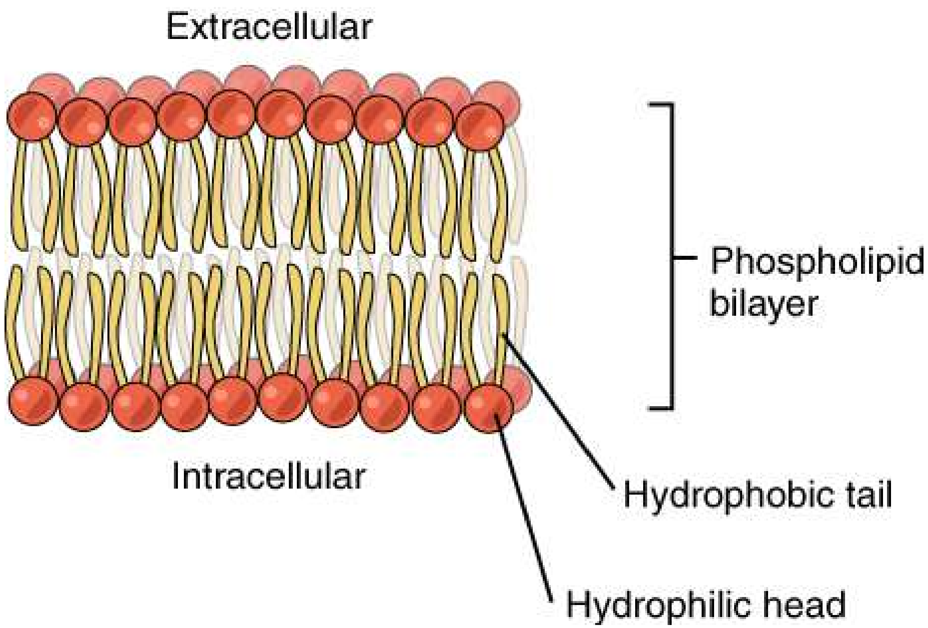
Hydrophilic head (attracted to water)

Hydrophobic fatty acid tails (repel water, attracted to other lipids: **lipophilic**)

A phospholipid with a hydrophilic head and a hydrophobic tail

Figure: https://commons.wikimedia.org/wiki/File:Phospholipid_Chemicalmakeup.png

- When placed in water, hydrophilic heads have no trouble, but hydrophobic tails clump together
  - – Cells have watery insides AND watery outsides, so phospholipid tails clump in such a way that 2 layers are formed: **phospholipid bilayer**



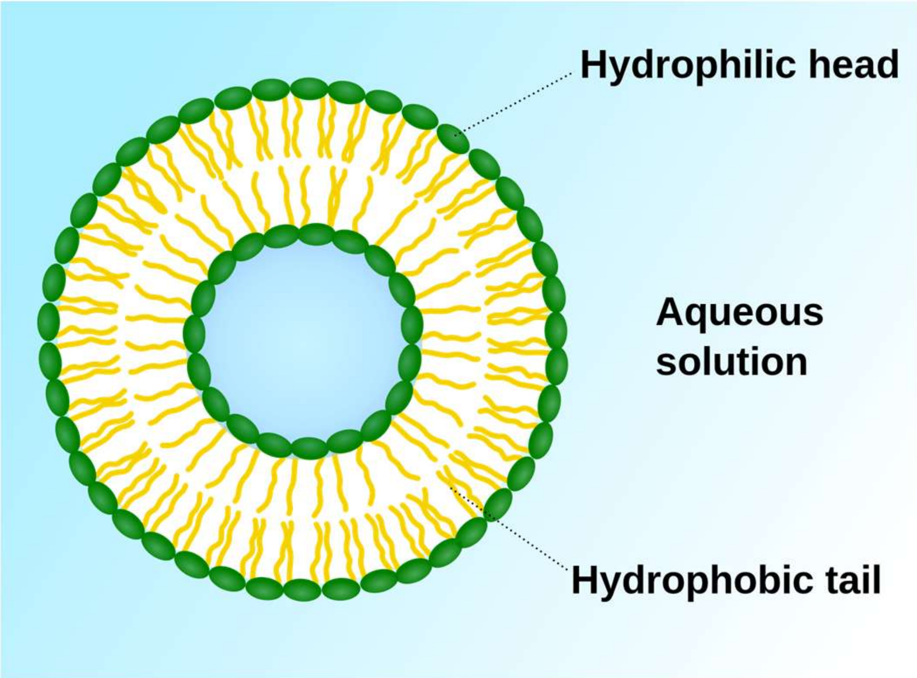


Figure: https://commons.wikimedia.org/wiki/File:0302_Phospholipid_Bilayer.jpg

Figure: https://commons.wikimedia.org/wiki/File:Liposome_scheme-en.svg

- Phospholipids are not bound together – they only take their shape because of the clumping of the tails

- The resulting phospholipid bilayer functions to <u>enclose the cell</u>
  - the hydrophilic & hydrophobic layers restrict movement, keeping the insides in & the outsides out

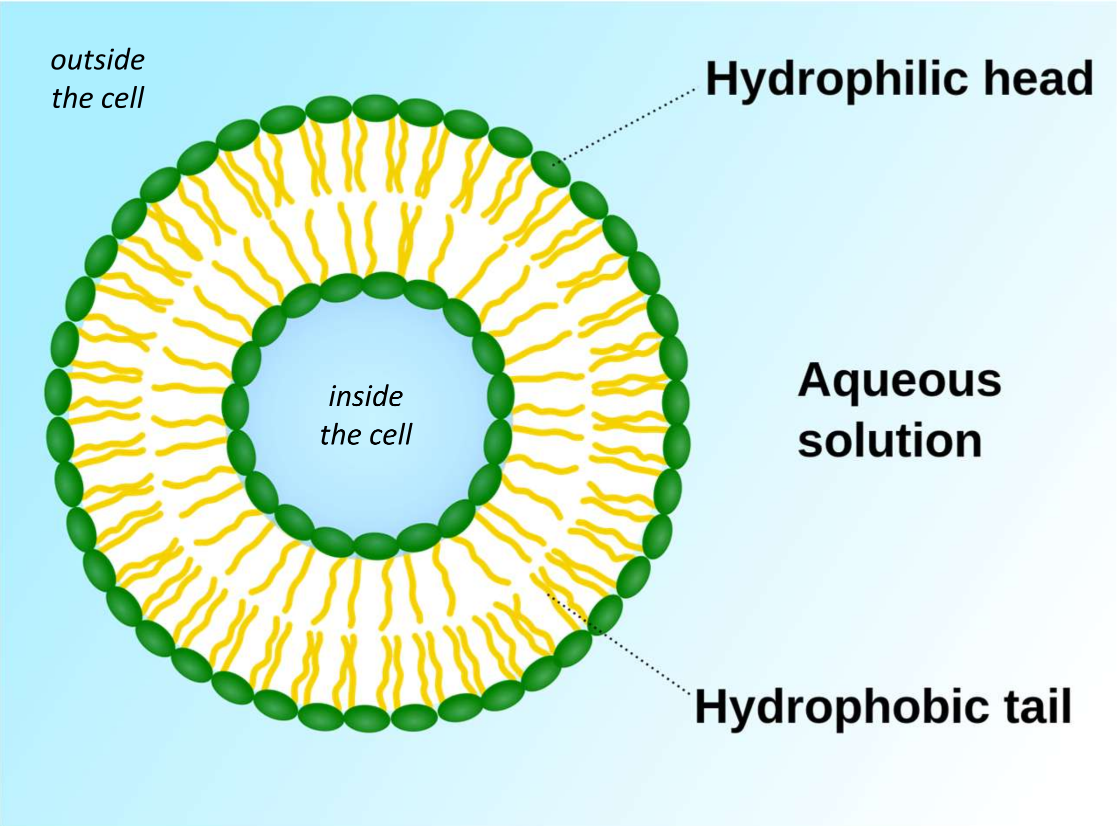
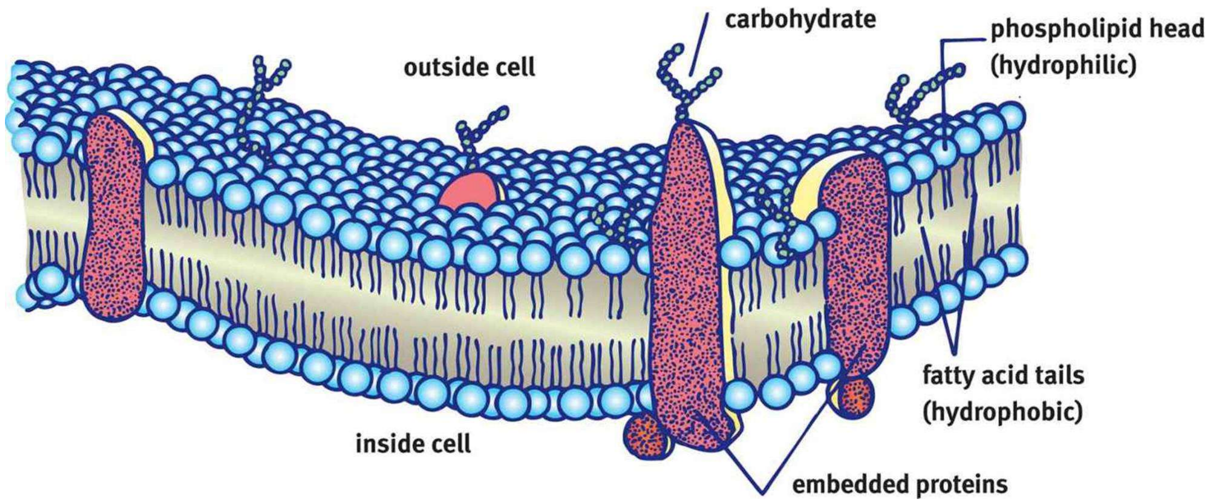
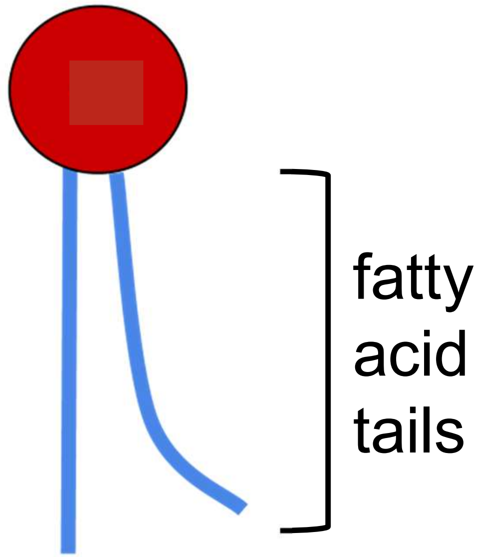


Figure: https://commons.wikimedia.org/wiki/File:Liposome_scheme-en.svg

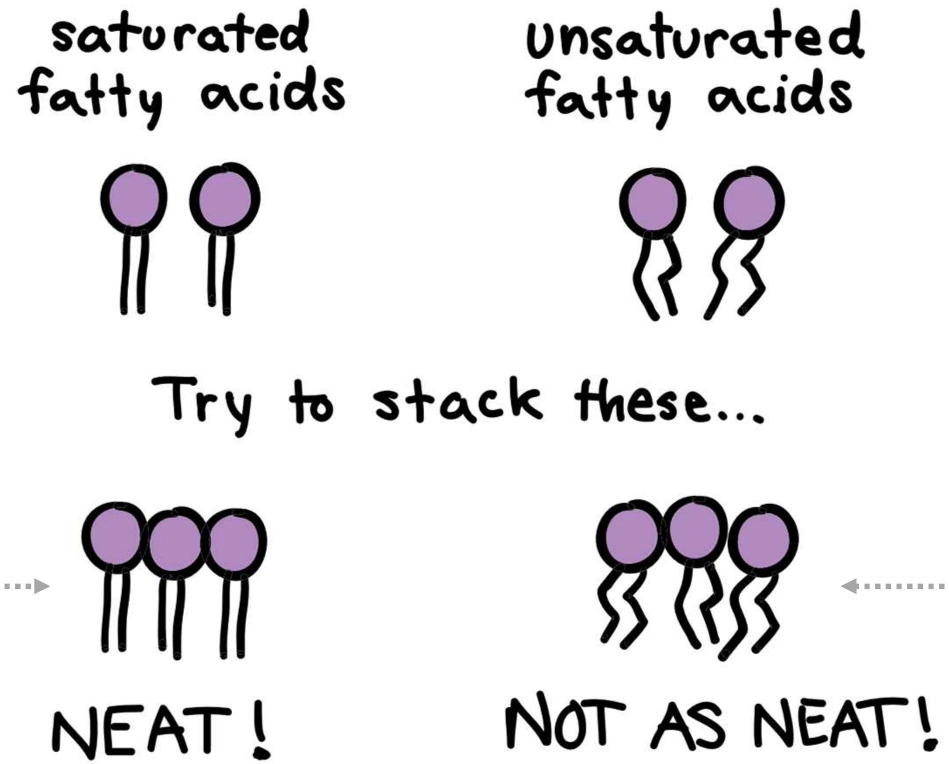
- Because phospholipids are not bound together, the cell membrane is very flexible or “fluid” rather than a rigid, unmoving straight line



- The fatty acid tails of the phospholipids can alter how fluid or flexible the membrane is



- Remember: **saturated** fatty acids stack well next to each other – this makes phospholipids <u>more stiff</u>
- Remember: **unsaturated** fatty acids don't line up well – this makes phospholipids <u>more flexible/fluid</u>

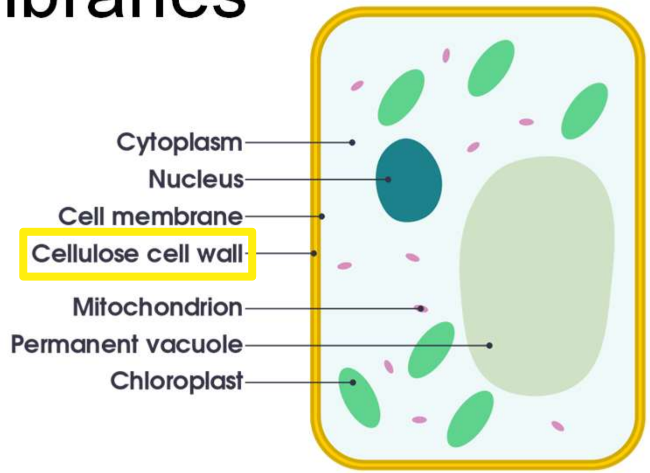


stiff/less fluid

flexible/more fluid

Figure: https://www.khanacademy.org/science/ap-biology/cell-structure-and-function/membrane-permeability/a/fluid-mosaic-model-cell-membranes-article

- Remember: plants make unsaturated fats
  - they have very flexible cell membranes
    - How do they stay rigid then?
      - Remember: they have cell walls for support



- Remember: animals don't have cell walls
  - How do they maintain structure then?
    - They make saturated fats to make their cell membranes more rigid & stiff
      - *Animals have 1 more thing that plants don't to help make their cell membranes more stiff…*

Figure: https://commons.wikimedia.org/wiki/File:Simple_diagram_of_plant_cell_(en).svg

- Component 2: **Cholesterol**
  - Remember: cholesterol is a hydrophobic lipid that animals make
    - wiggles in with the hydrophobic tails to make the cell membrane more stiff

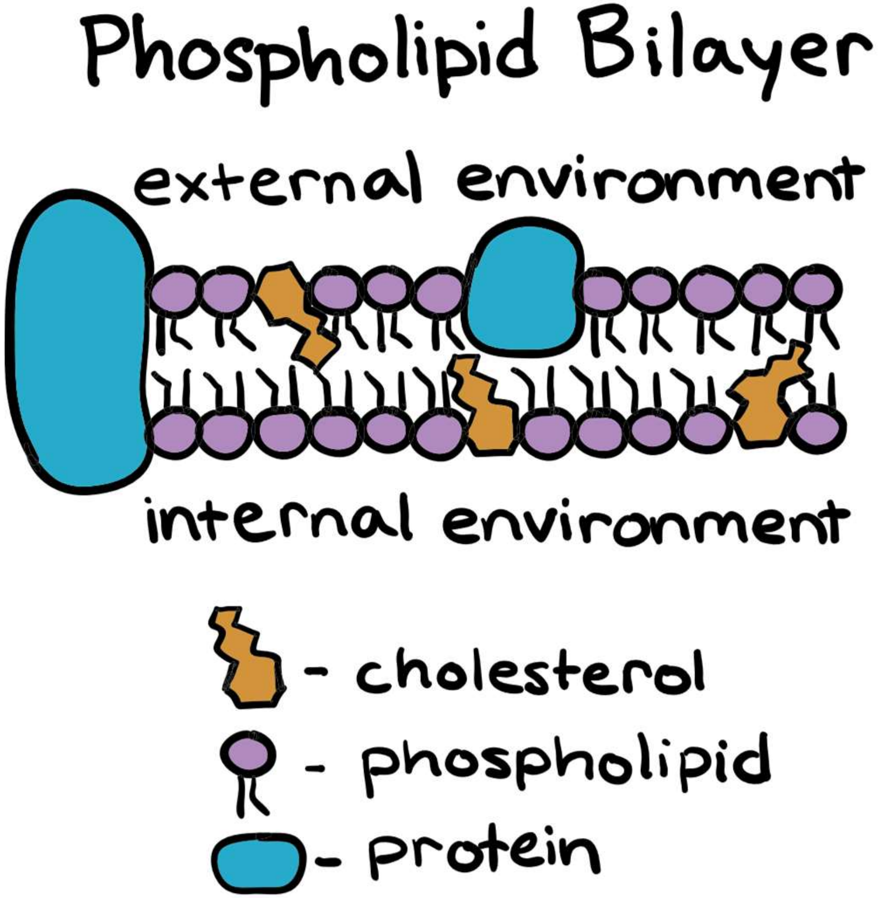


Figure: https://www.khanacademy.org/science/ap-biology/cell-structure-and-function/membrane-permeability/a/fluid-mosaic-model-cell-membranes-article

- Recap: cell membranes are fluid & flexible because the phospholipids that make them up are not bound to one another
  – we can increase fluidity (make it more flexible)
    • use unsaturated fatty acids
    • use no or very little cholesterol
  – or we can decrease fluidity (make it more stiff)
    • use saturated fatty acids
    • use plenty of cholesterol

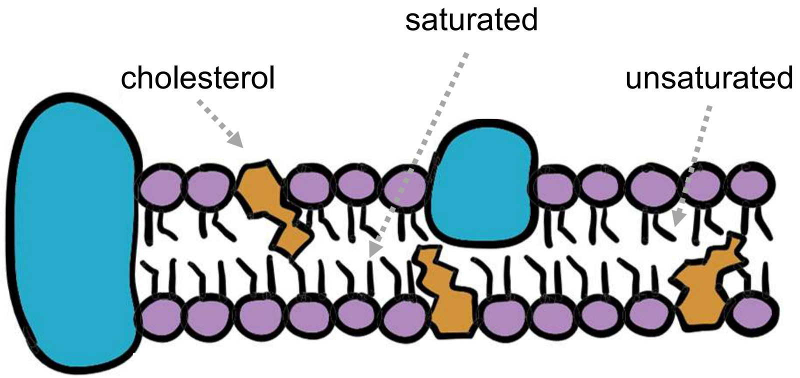


Figure: https://www.khanacademy.org/science/ap-biology/cell-structure-and-function/membrane-permeability/a/fluid-mosaic-model-cell-membranes-article

What molecules can & can not cross cell membranes on their own?

- Even though the phospholipid bilayer provides a barrier, some substances can still get across
  - Very <u>small molecules</u> like water & gases (CO_2 & O_2) can wiggle though the phospholipids

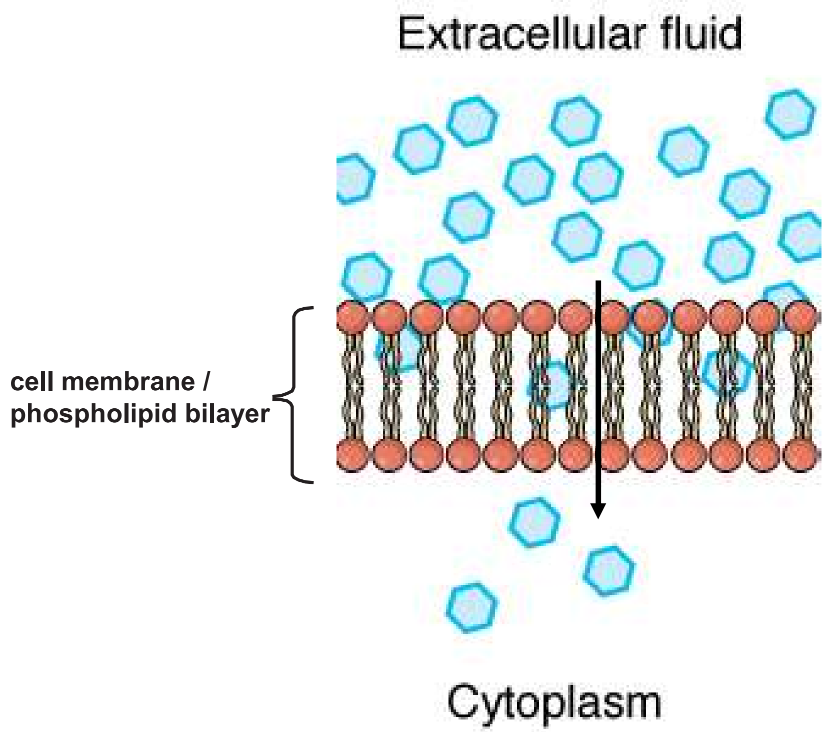
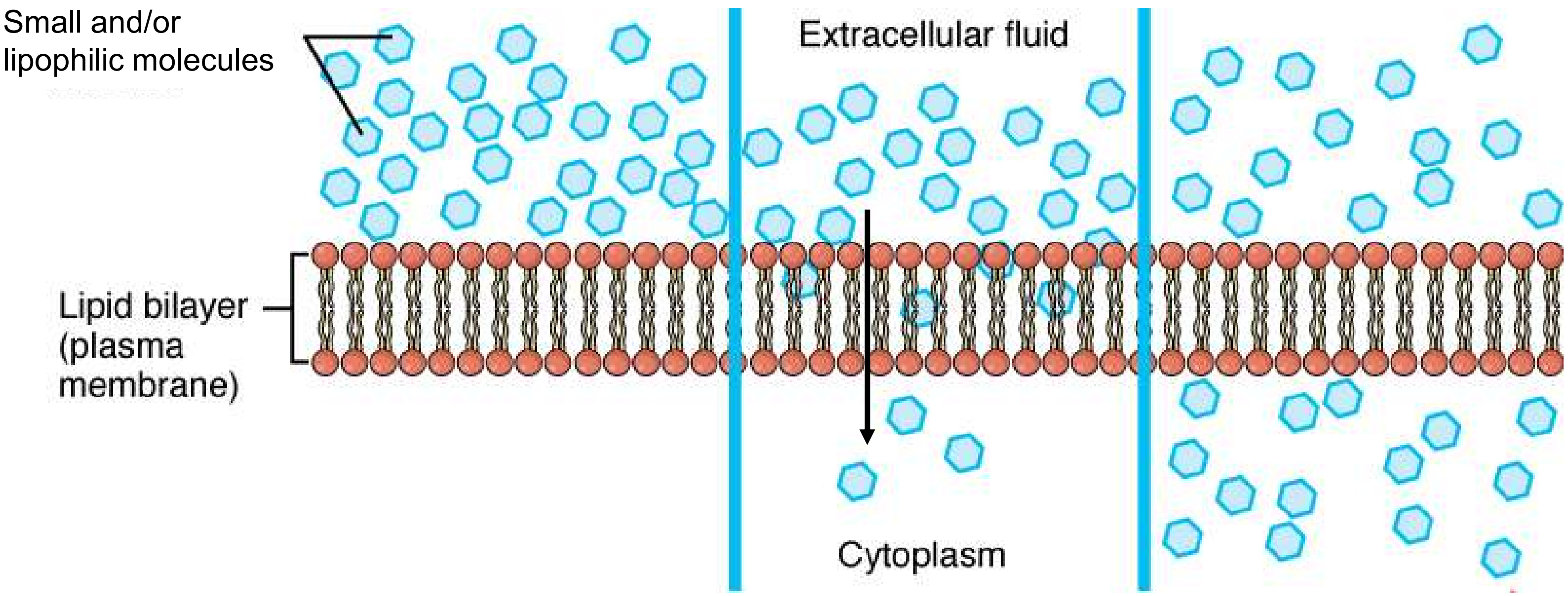


Figure: https://commons.wikimedia.org/wiki/File:0305_Simple_Diffusion_Across_Plasma_Membrane.jpg

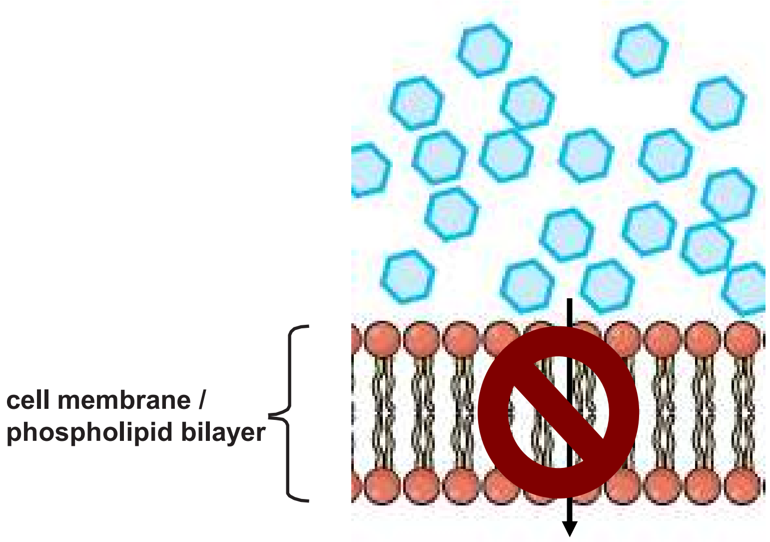
- Can also pass: hydrophobic molecules – also called "**lipophilic**" molecules
  - Lipophilic molecules are lipid-loving, so they are able to move easily through the fatty acid layer



Caffeine, alcohol, hormones, & some vitamins are lipophilic (hydrophobic)

Figure: https://commons.wikimedia.org/wiki/File:0305_Simple_Diffusion_Across_Plasma_Membrane.jpg

- Even though some things can cross, the bilayer prevents many substances from crossing



  - <u>Hydrophilic molecules</u> can NOT cross the phospholipid bilayer
    - e.g. salts & sugars
  - Very <u>large molecules</u> also can NOT cross
    - e.g. polysaccharides

- However: we need to get salts, sugars, polysaccharides, & other large or hydrophilic molecules inside our cells in order to stay alive
  - *Remember: life requires energy & nutrients*

- So how do we get these things across the cell membrane?
  - Using the 3rd component of membranes: **proteins**

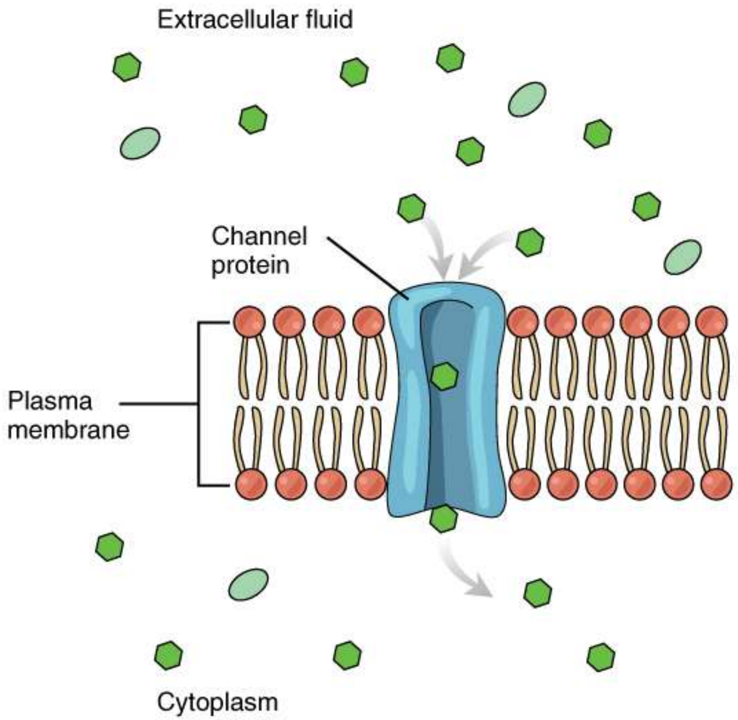


Figure: https://commons.wikimedia.org/wiki/File:0306_Facilitated_Diffusion.jpg

- Component 3: **Proteins**
  - There are 5 types of proteins associated with the cell membrane: each has a different function

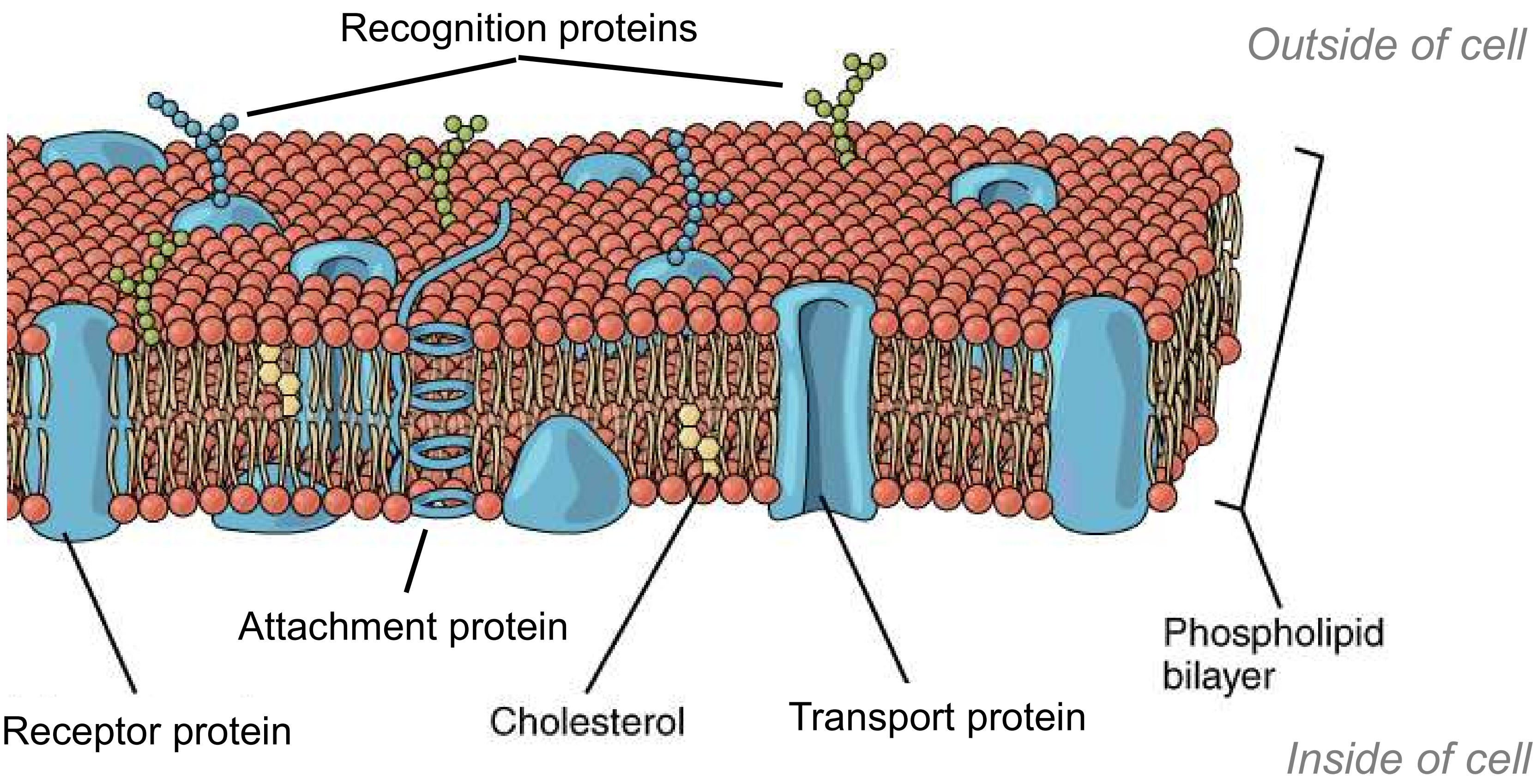
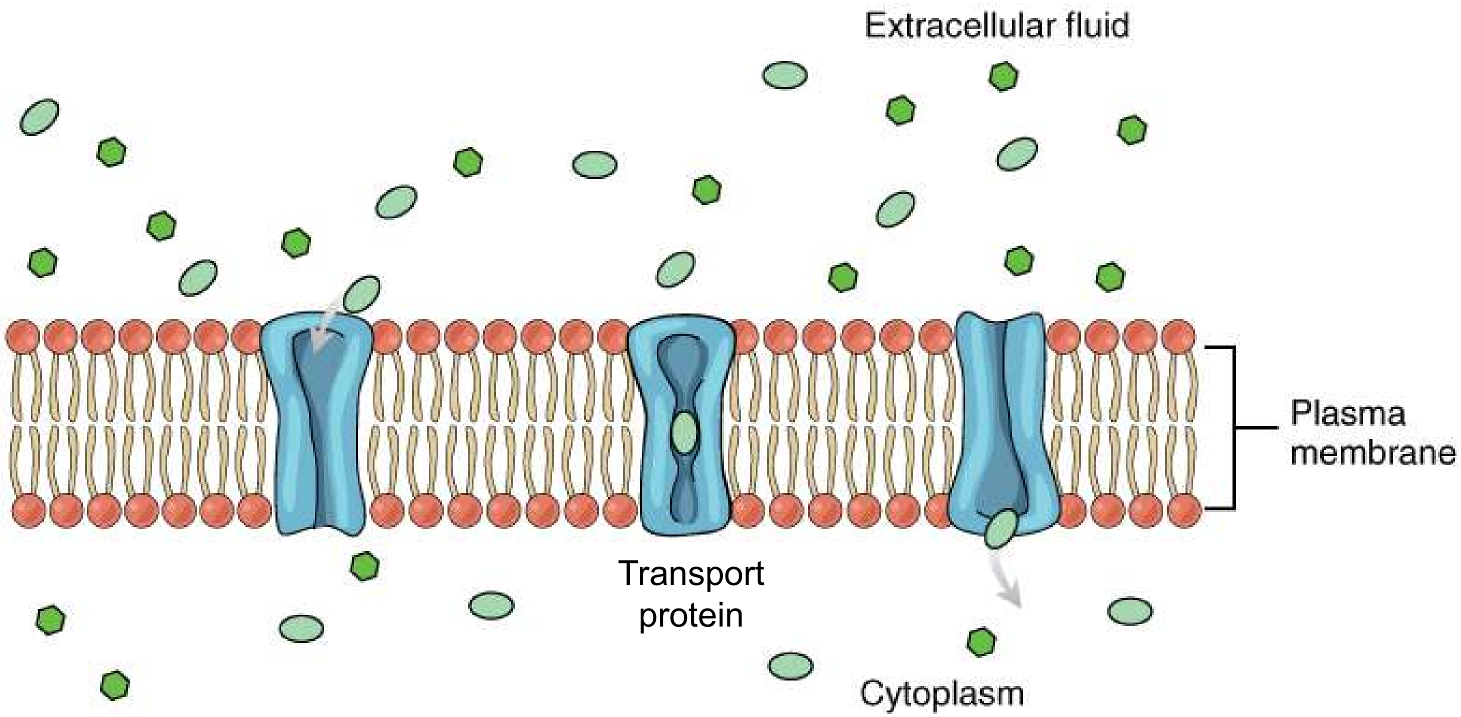


Figure: https://commons.wikimedia.org/wiki/File:0303_Lipid_Bilayer_With_Various_Components.jpg

- Type 1: **transport proteins**
  - Allow specific molecules to cross the membrane
  - e.g. a sodium transport protein ONLY lets sodium through



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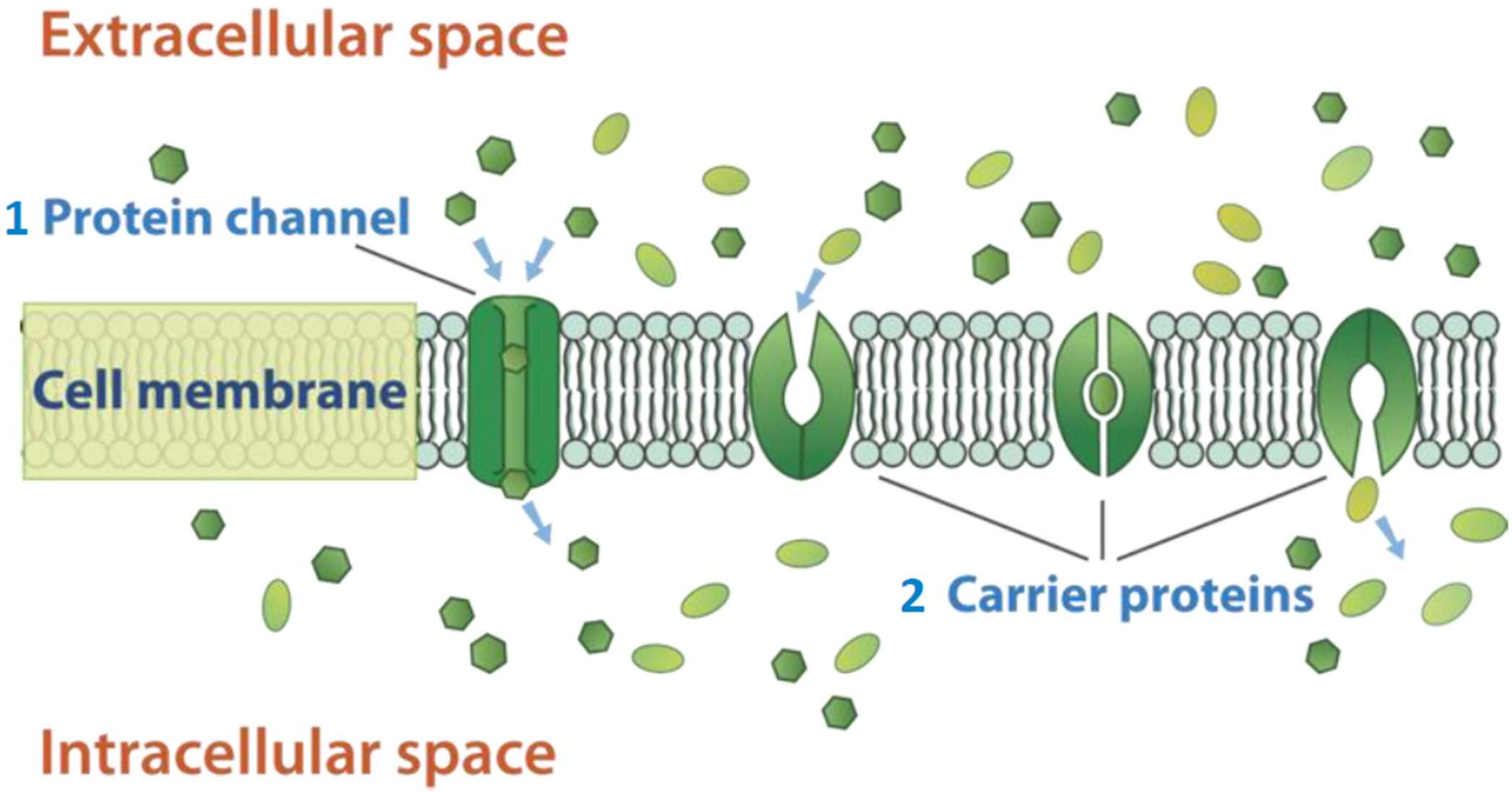

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- Type 2: **recognition proteins**
  - Serve as ID tags on the surface of a cell - this prevents the immune system from attacking it

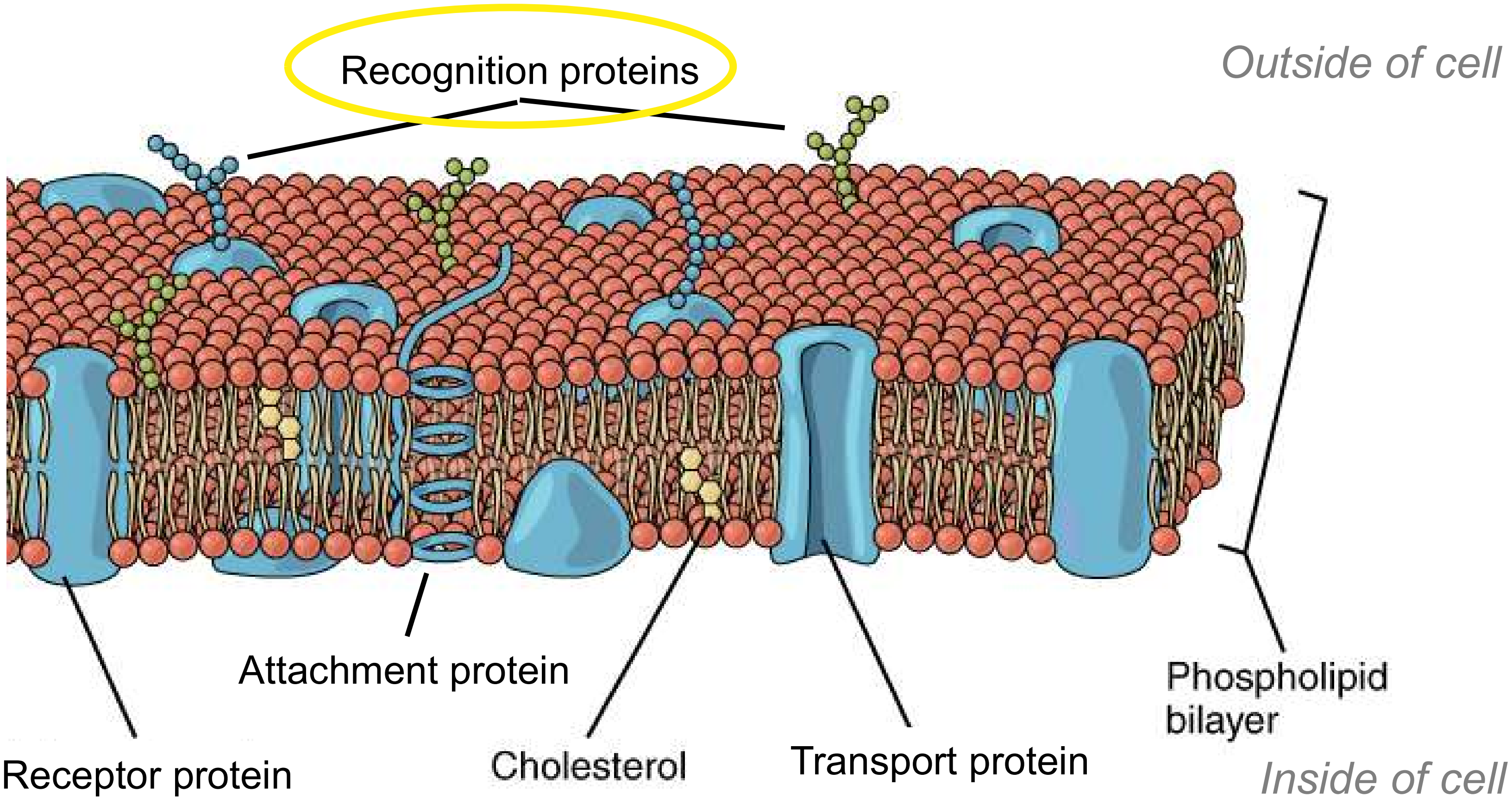
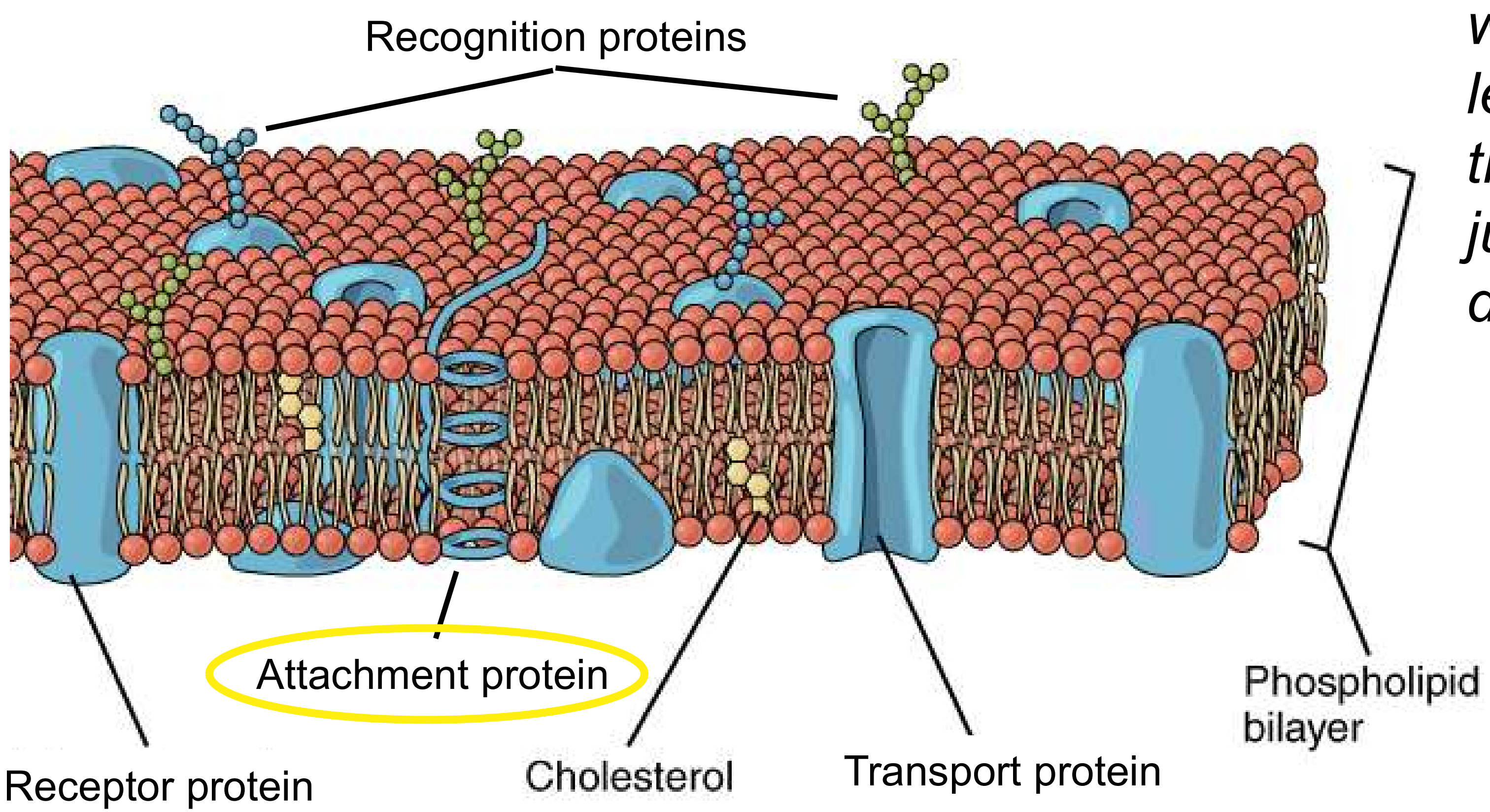


Figure: https://commons.wikimedia.org/wiki/File:0303_Lipid_Bilayer_With_Various_Components.jpg

- Type 3: **attachment proteins**
  - Anchor & attach the cell membrane to the inner cytoskeleton & to other cells



Remember: we already learned about these - tight junctions & desmosomes

Figure: https://commons.wikimedia.org/wiki/File:0303_Lipid_Bilayer_With_Various_Components.jpg

- Type 4: **enzymes**
  - Regulate chemical reactions
  - Not all enzymes are associated with the cell membrane
    - *more in the next chapter*

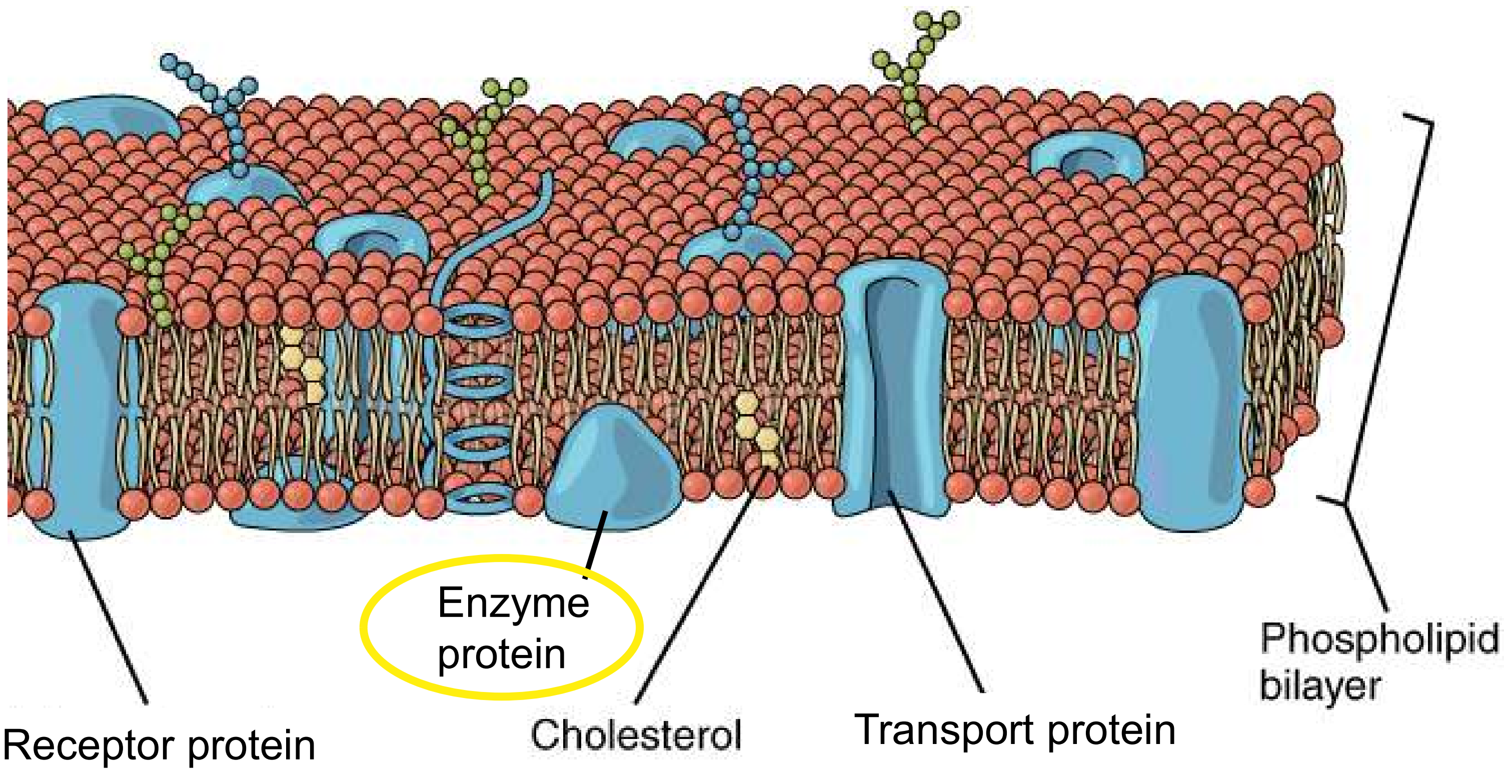
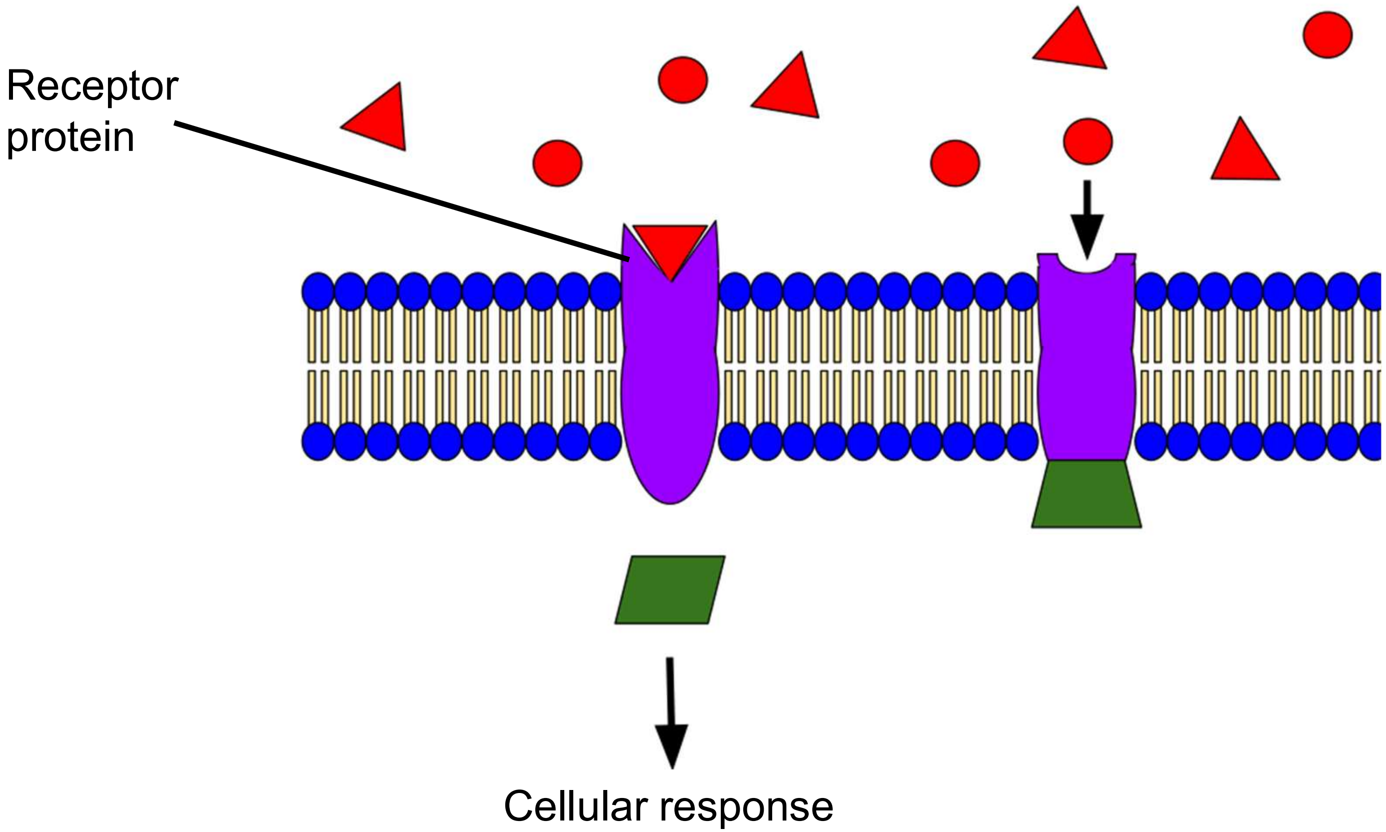


Figure: https://commons.wikimedia.org/wiki/File:0303_Lipid_Bilayer_With_Various_Components.jpg

- Type 5: **receptor proteins**
  - Trigger cell responses when they bind to a specific molecule



- Receptor proteins trigger cell responses when they bind to a <u>specific</u> molecule
  - e.g. an insulin receptor will ONLY respond to the hormone insulin
  - molecule triggers a change in the cell, but never actually enters the cell
    - *molecule can now detach & go on to communicate with many other cells*

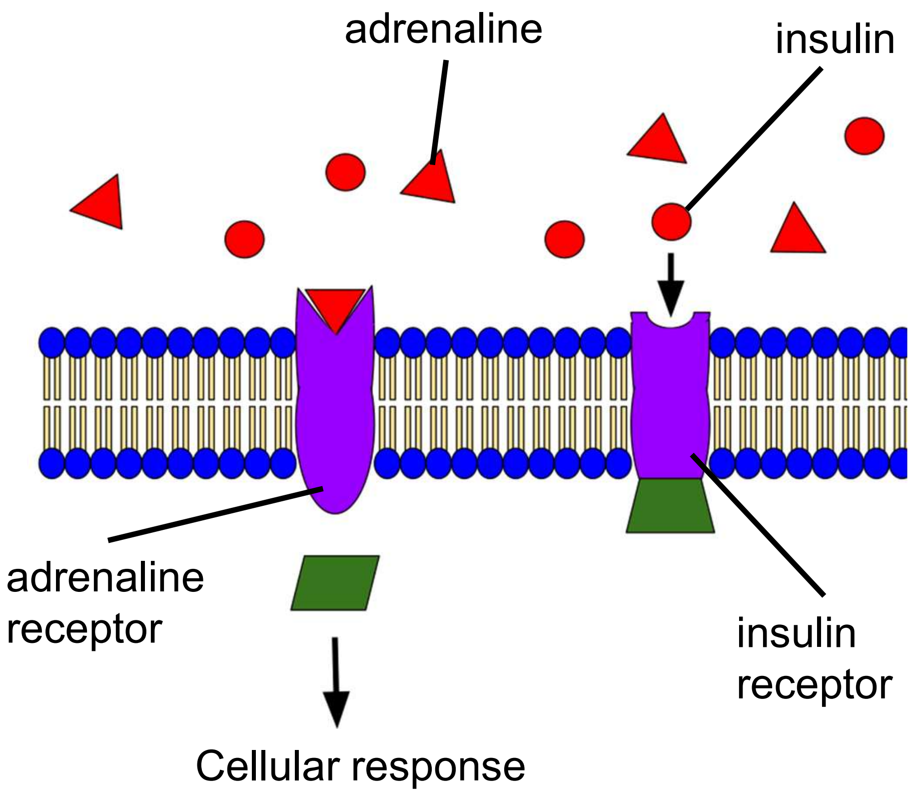


Figure: https://commons.wikimedia.org/wiki/File:Receptor_(Biochemistry).svg

- We’ve now seen that there are several components to cell membranes, each with their own function
  - *Phospholipids enclose cells*
  - *Cholesterol makes cell membranes more rigid*
  - *5 proteins have 5 different functions*

- We also know some substances can move across the phospholipid bilayer by themselves, & some need transport proteins to get across
  - <u>So how do any of these molecules know whether to enter the cell, or to exit the cell?</u>

How do molecules know whether to move in or out of a cell?

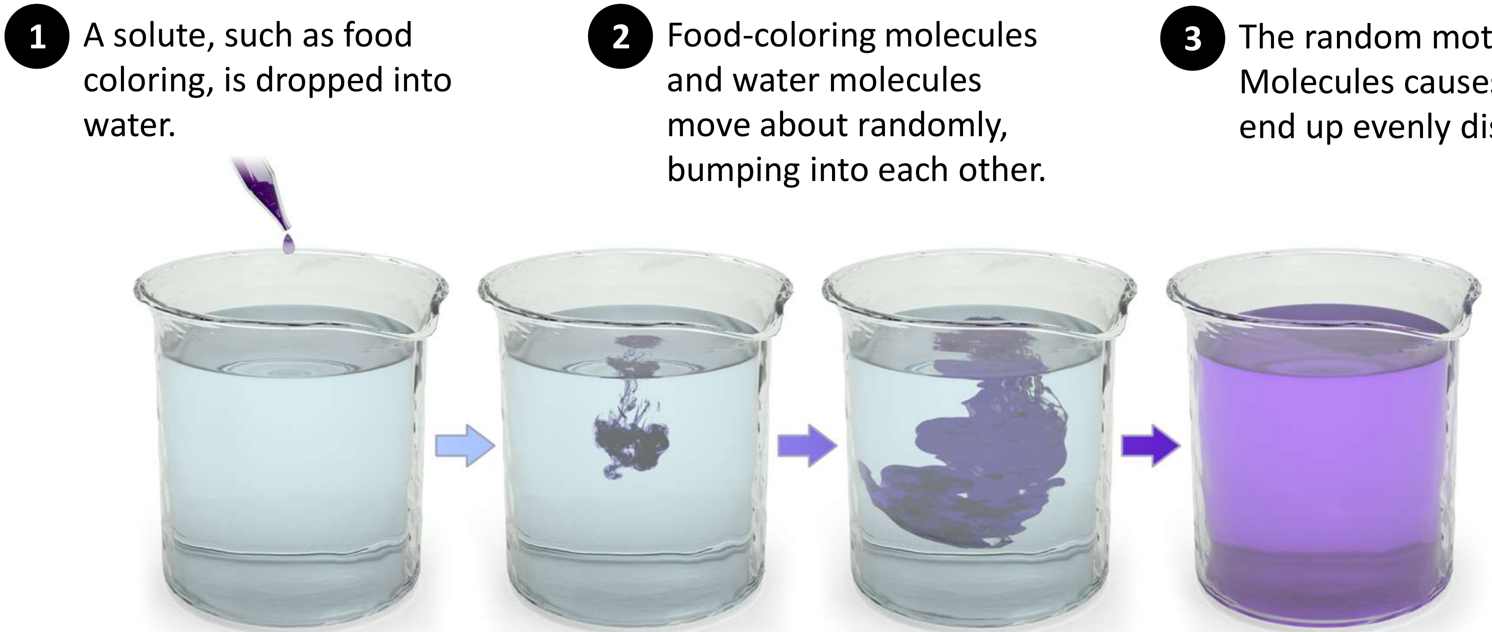
- Technically, molecules don't "know" anything – but they reliably move according to the laws of physics
  - All molecules dissolved in water, or "**solutes,**" are in constant, random motion

dye molecules are solutes

water molecules AND dye molecules are constantly moving around in the beaker

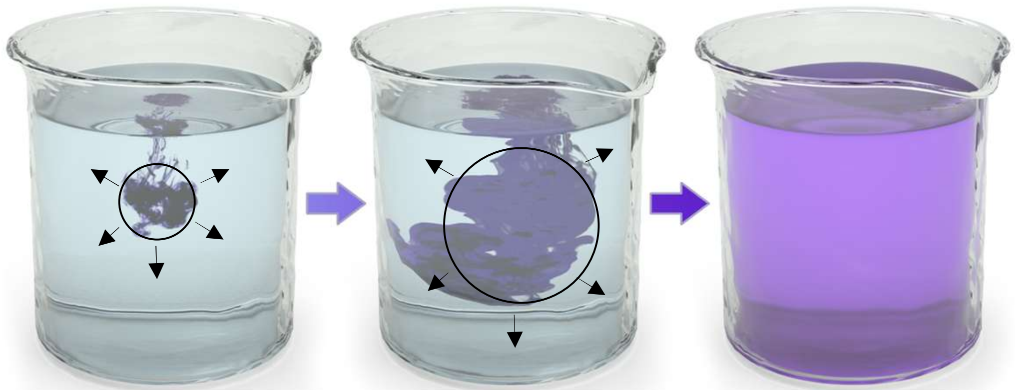
Figure: https://commons.wikimedia.org/wiki/File:Blausen_0315_Diffusion.png

- Even though solutes are moving randomly, we can statistically predict how they will move overall
  - We see here that even though solutes are moving randomly, they eventually become evenly distributed



- In other words, solutes are statistically more likely to move from where they are more highly concentrated to where there is a lower concentration
  - this random movement from high to low concentration = movement "down the **concentration gradient**"
  - eventually, this leads to **equilibrium**: an even distribution of solutes & water molecules

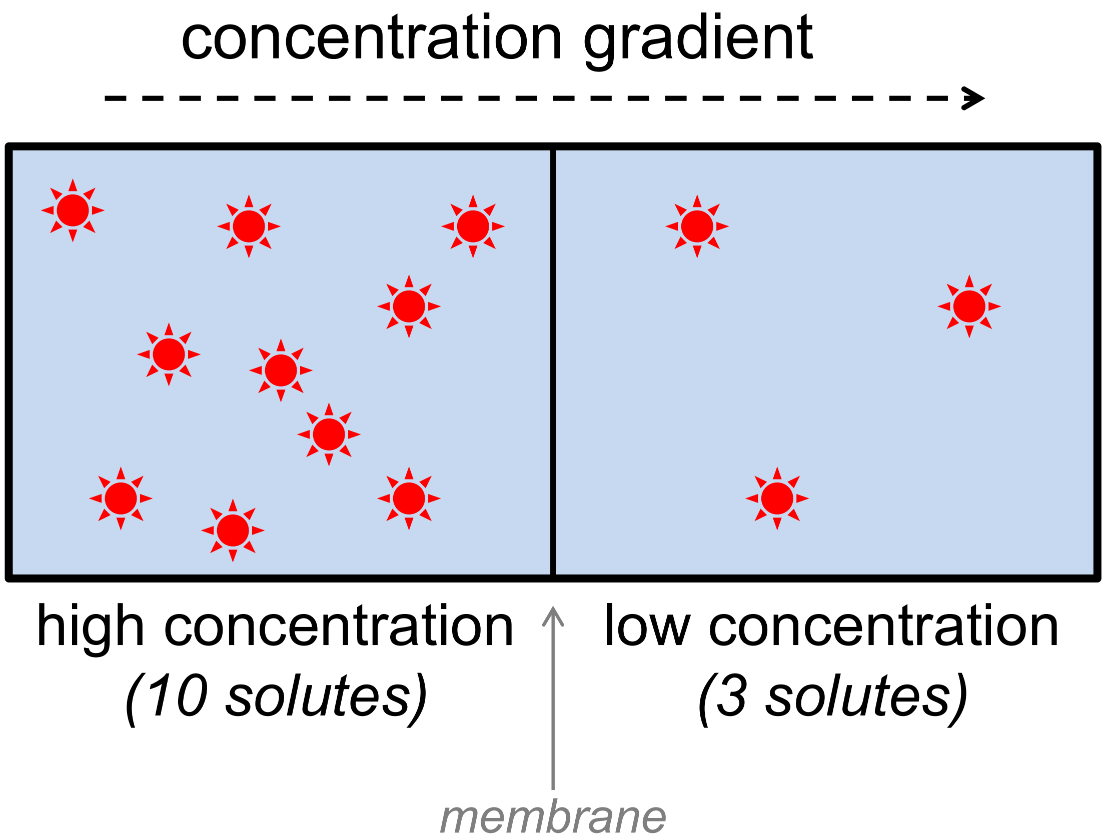
Dye is moving down the concentration gradient, with molecules moving from high to low concentration



Water and dye molecules have reached equilibrium

Figure: https://commons.wikimedia.org/wiki/File:Blausen_0315_Diffusion.png

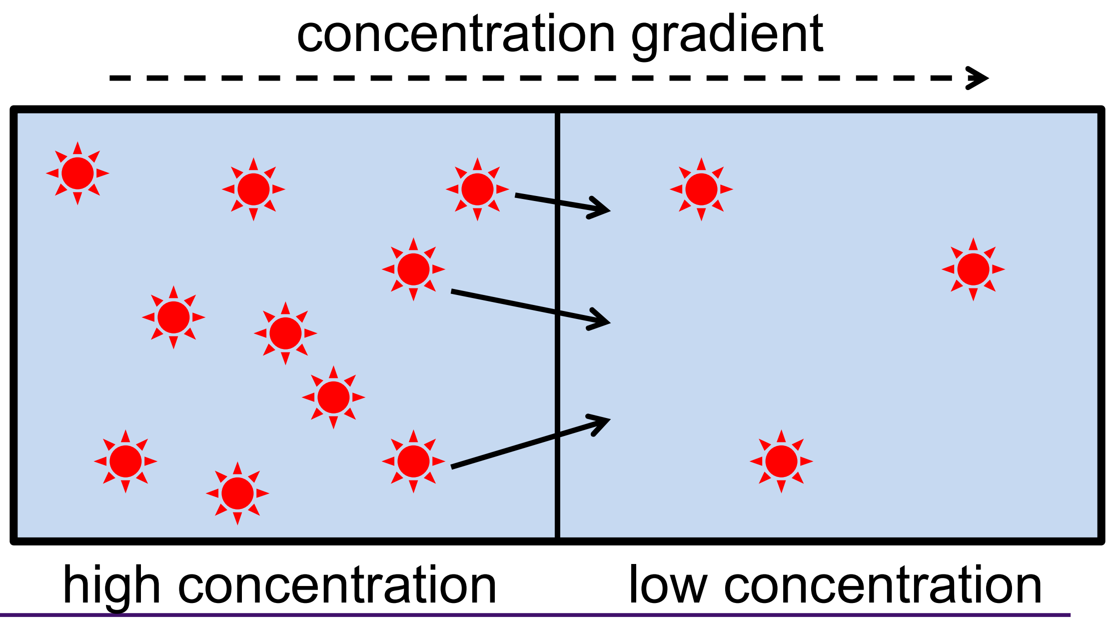
- Now let's consider what would happen if there were a membrane between 2 solutions:



10 solutes might move from left to right, but only 3 solutes might move in the opposite direction: this means statistically there will be a <u>net movement</u> of solutes from left to right, <u>down the concentration gradient</u>

- Since molecules move down their concentration gradient, we can now understand:
  - molecules will move IN to a cell if there is a lower concentration inside
  - molecules will move OUT of a cell if there is a lower concentration outside

- The movement of molecules down their concentration gradient = **diffusion**



What are the possible ways to move across a cell membrane?

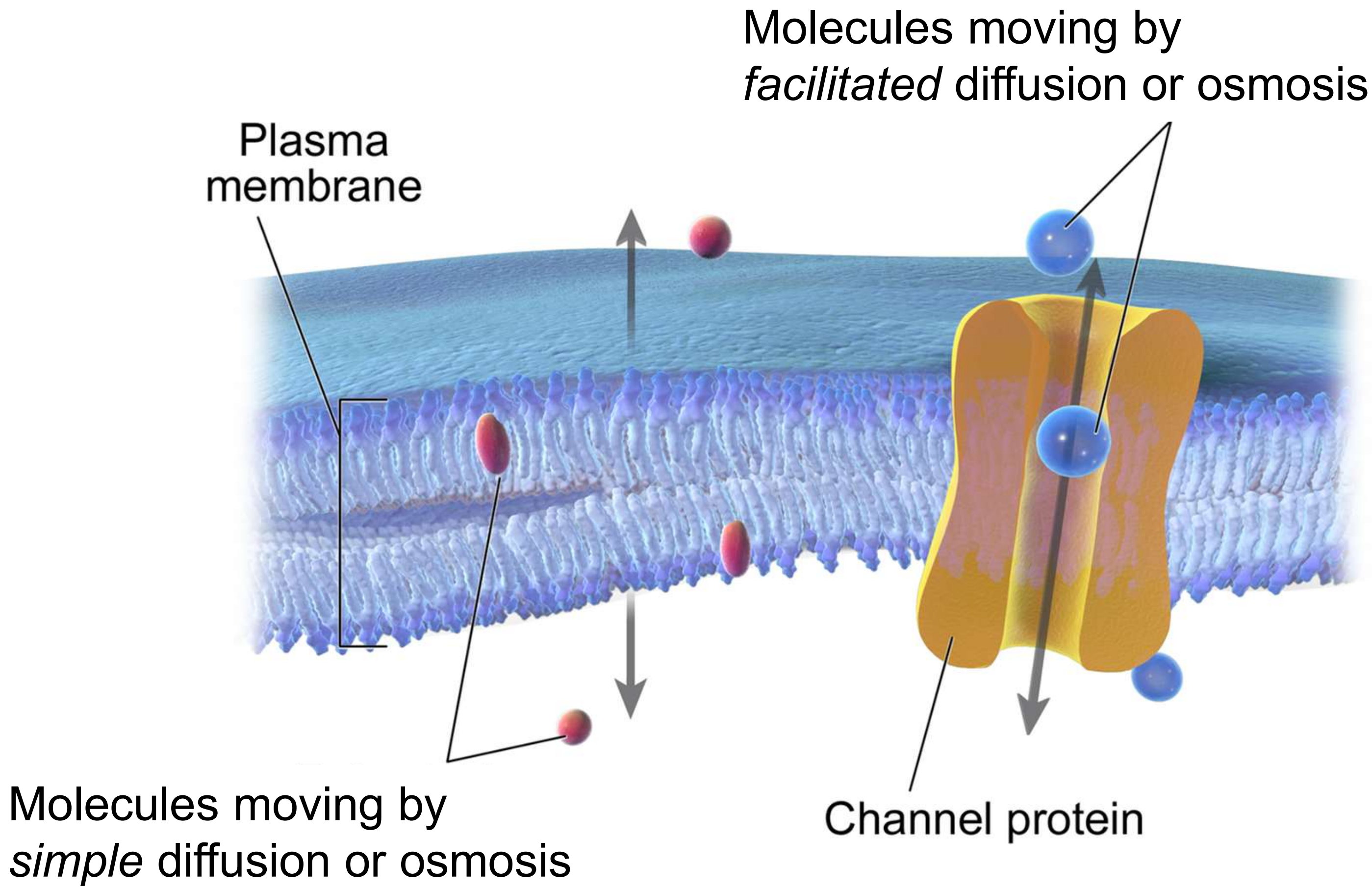
- There are 6 types of transport across cell membranes
  - 3 of them are called **passive transport**
    - Substances move down their concentration gradient (high to low)
    - No energy is needed
  - 3 of them are called **energy-requiring transport**
    - Energy is required

- **Passive transport** occurs when molecules move across a membrane without energy – they move down their concentration gradient

High concentration

↓ *across a membrane*

Low concentration



- **Simple diffusion**
  - Remember: small molecules & lipophilic molecules can pass through the phospholipid bilayer on their own
    - they move directly across the membrane from high to low concentration
    - no energy is needed or used

= simple diffusion

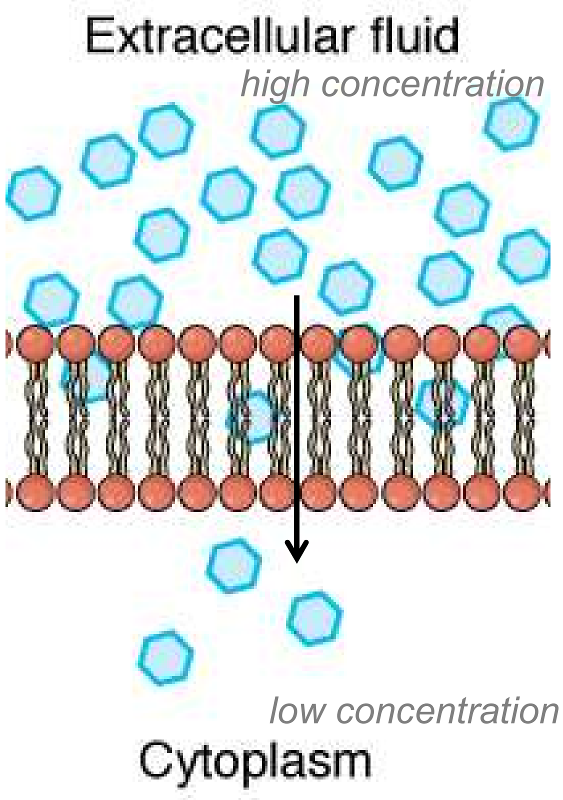


Figure: https://commons.wikimedia.org/wiki/File:0305_Simple_Diffusion_Across_Plasma_Membrane.jpg

- **Facilitated diffusion**
  - – Remember: hydrophilic molecules can NOT pass through the phospholipid bilayer on their own
    - they need a protein to help them across the membrane from high to low concentration
    - no energy is needed or used

= <u>facilitated diffusion</u>

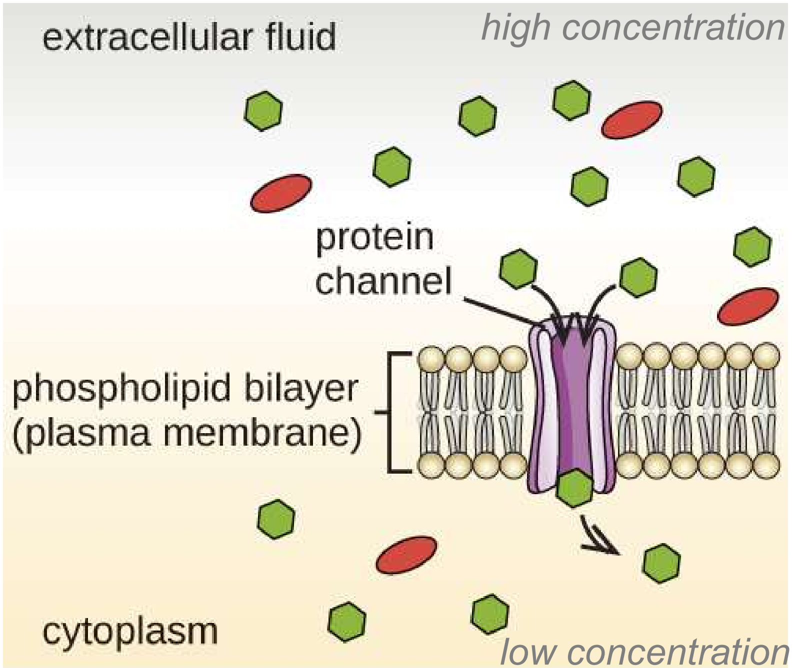
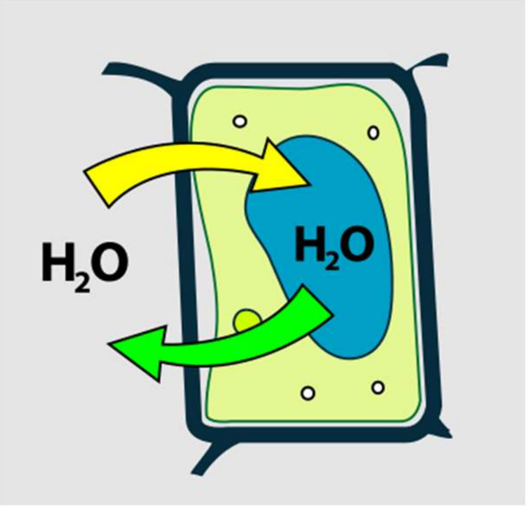


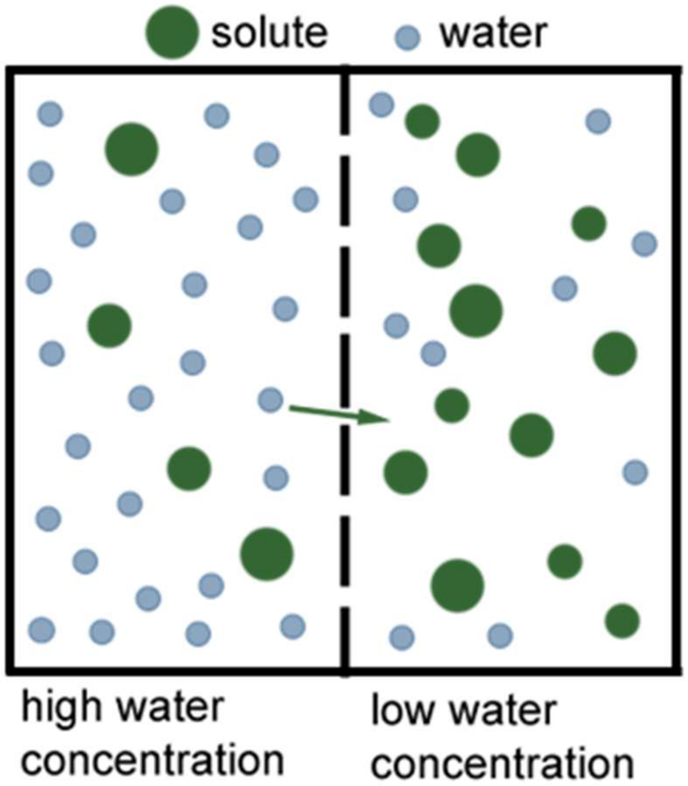
Figure: https://commons.wikimedia.org/wiki/File:OSC_Microbio_03_03_facdiff.jpg

- **Osmosis**
  - Sometimes there is a concentration gradient, but solutes can not move past the membrane
    - *maybe they are too big, or transport proteins are closed*
  - Water is small & can cross membranes: if solutes can't diffuse, water will move until each side of the membrane has equal concentrations (equilibrium)
    - **Osmosis** = movement of water across a membrane

OSMOSIS
Water molecules diffuse across a cell membrane until the concentration of water inside & outside the cell is equalized.



- **Osmosis:** water technically moves by diffusion: from a region of high to low water concentration

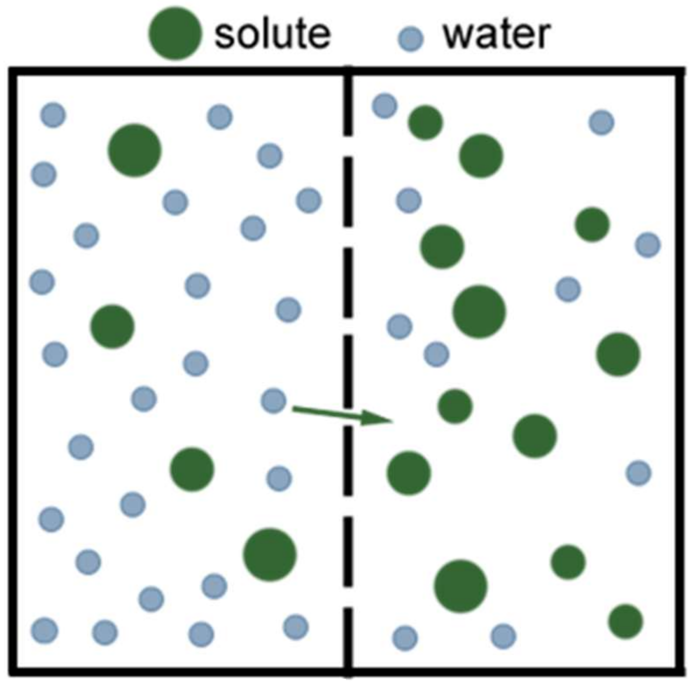


net movement of osmosis to the right

(water moves from high to low water)

Figure: https://scienceaid.net/biology/cell/osmosis.html

- However, it can be hard to determine which area has high vs. low water
  - to figure out which way water will move (osmosis) to help reach equilibrium, find the side that has high solutes – those must be diluted out to be closer to the concentration on the other side, so water will move toward them



5% solute
(lower solute concentration)

10% solute
(higher solute concentration)

net osmosis to the right
(water moves to dilute out the most concentrated solute)

Figure: https://scienceaid.net/biology/cell/osmosis.html

- The drive for equilibrium is so powerful that osmosis overcomes gravity:

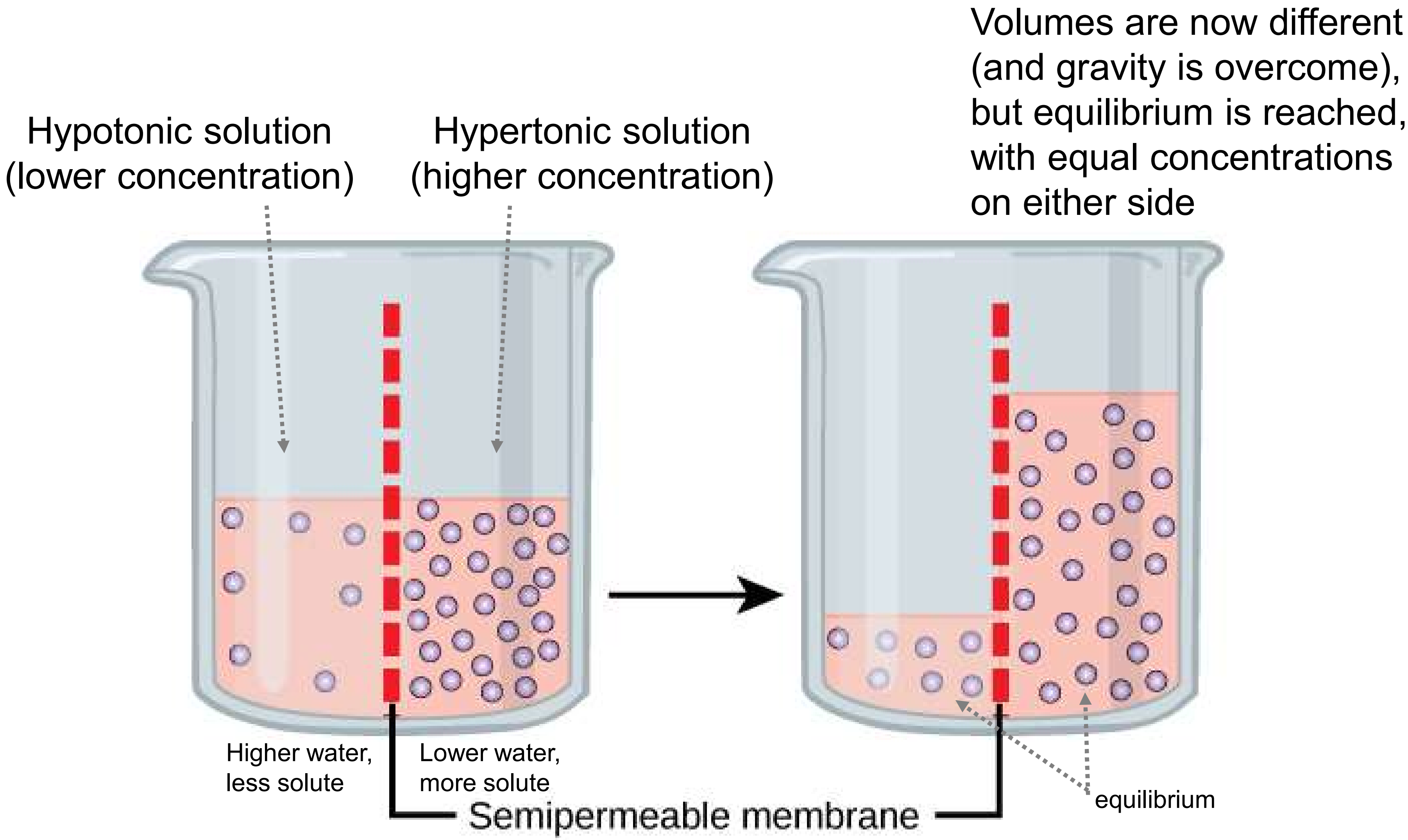
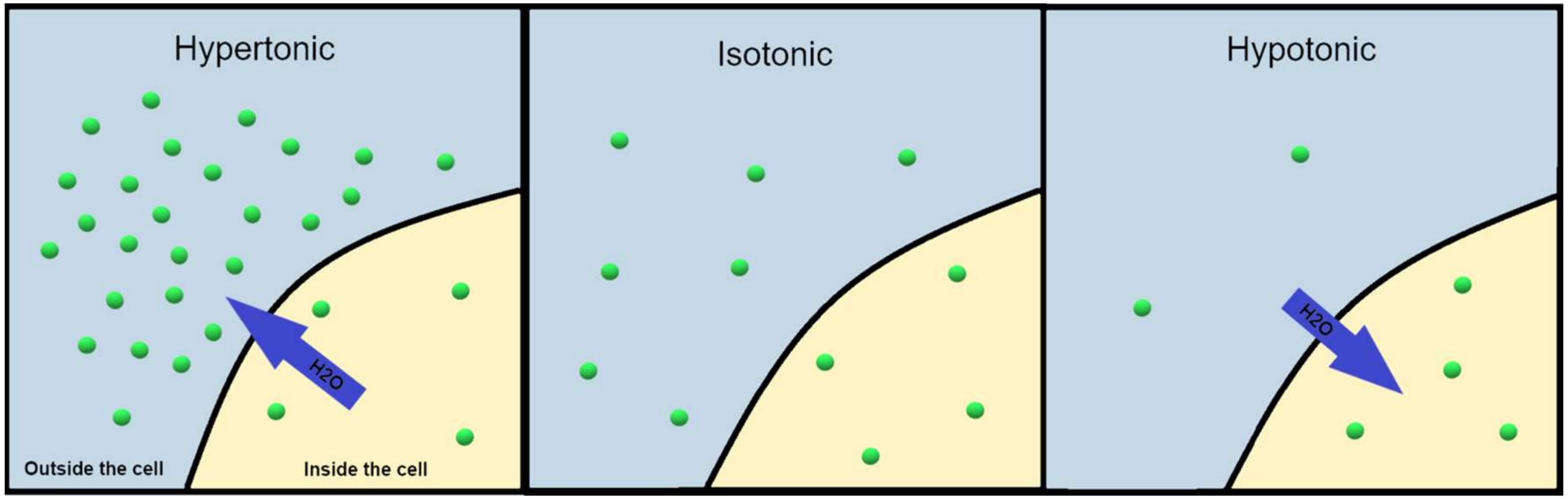


Figure: https://commons.wikimedia.org/wiki/File:Figure_05_02_06.jpg

- Cells are constantly filled with & surrounded by water: equilibrium is crucial, or life stops
  – Our cells can find themselves in 3 possible environments: isotonic, hypertonic & hypotonic

equilibrium



A cell in a **hypertonic** solution loses water and shrinks

A cell in an **isotonic** solution remains healthy

A cell in a **hypotonic** solution gains water and can swell and burst

Figure: https://commons.wikimedia.org/wiki/File:Hypertonic_Isotonic_Hypotonic.png

- **Isotonic** solutions: equal concentrations of water & solutes on either side of the membrane
  - – No net water movement occurs

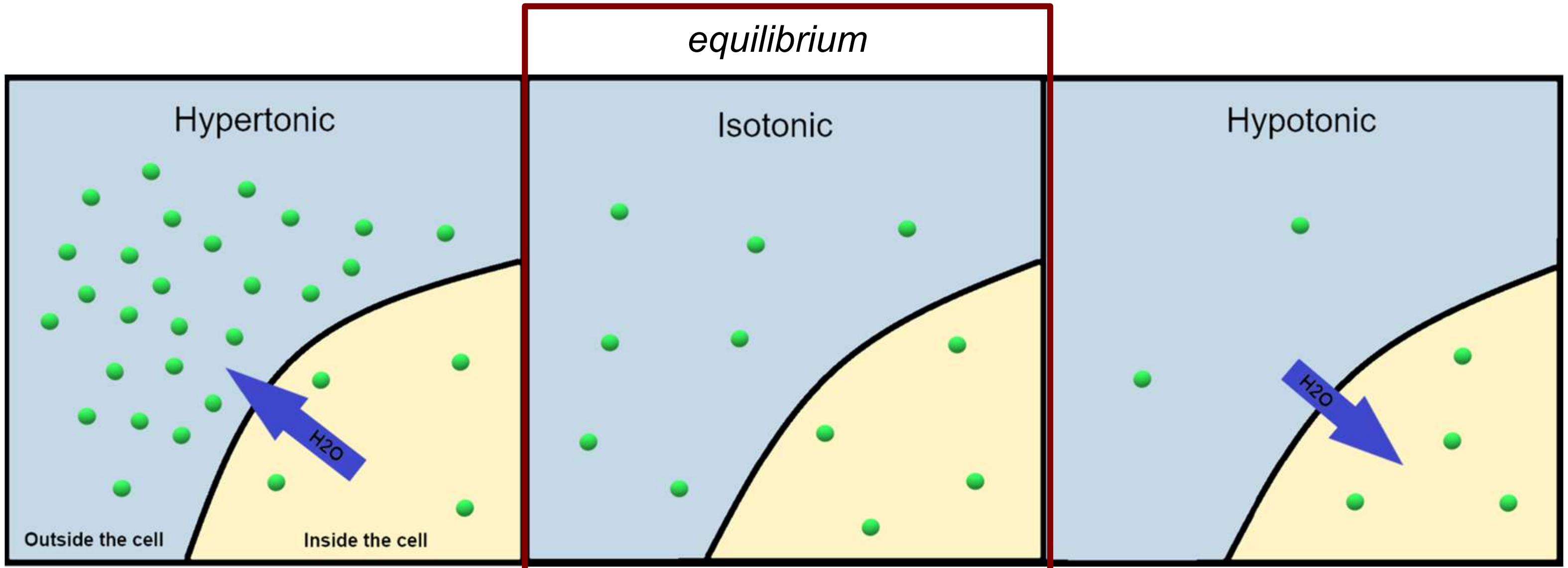


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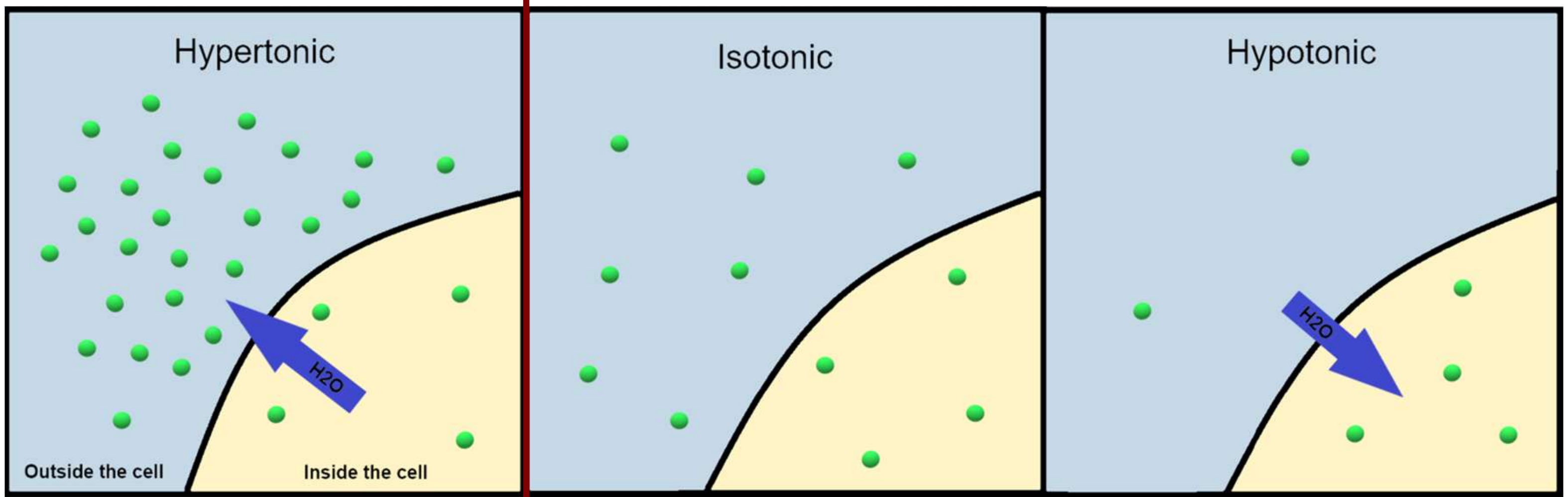
Figure: https://commons.wikimedia.org/wiki/File:Hypertonic_Isotonic_Hypotonic.png

- A **hypertonic** solution is one with a greater solute concentration relative to the cell
  - Water moves toward the more concentrated solutes outside the cell to dilute them: cell shrivels

equilibrium



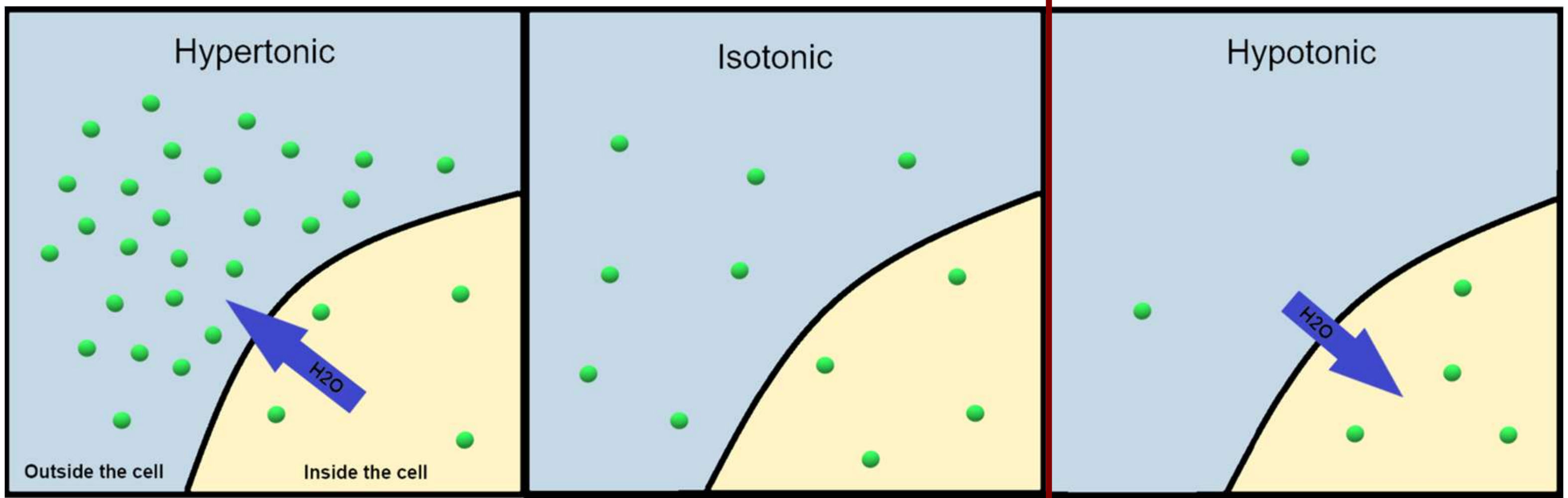
A cell in a **hypertonic** solution loses water and shrinks

A cell in an **isotonic** solution remains healthy

A cell in a **hypotonic** solution gains water and can swell and burst

- A **hypotonic** solution is one with a <u>lower solute</u> concentration relative to the cell
  - – Water moves toward the more concentrated solutes inside the cell to dilute them: cell swells

equilibrium



A cell in a **hypertonic** solution loses water and shrinks

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Figure: https://commons.wikimedia.org/wiki/File:Hypertonic_Isotonic_Hypotonic.png

- <u>How these solutions affect our cells:</u>
  - – Red blood cells are suspended in our watery blood
    - If our blood stream is **isotonic** relative to our cells, no net movement occurs – cells stay healthy

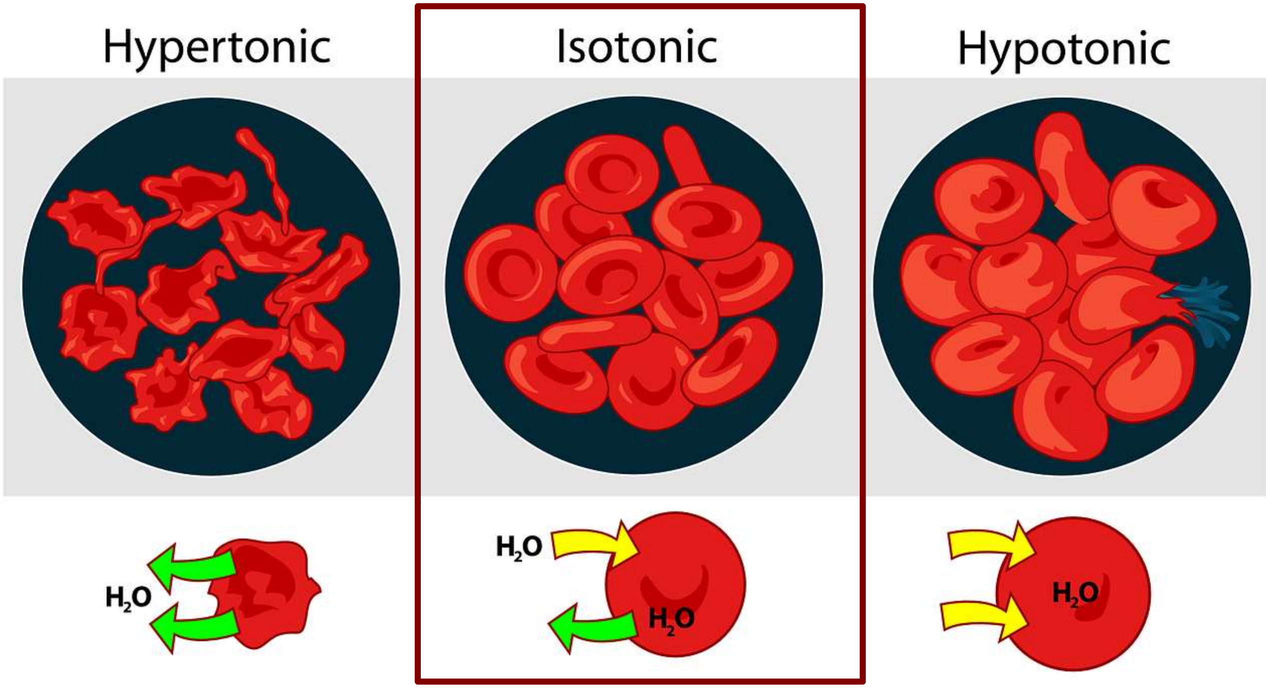
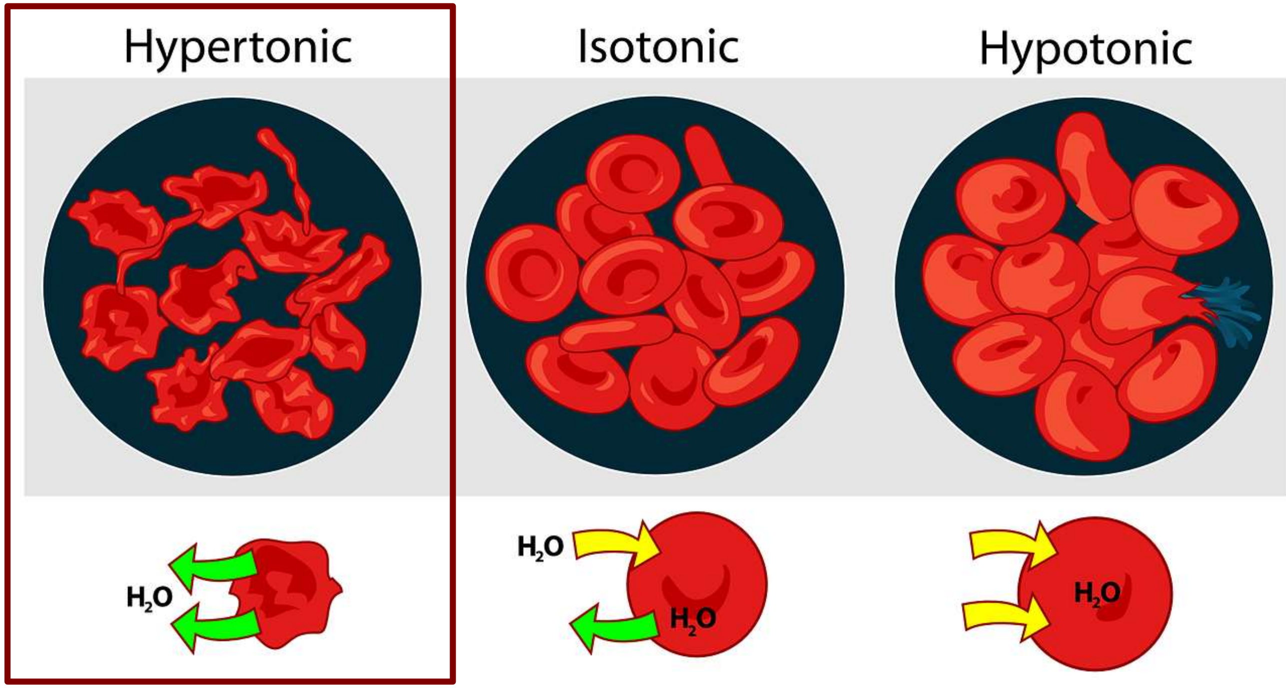


Figure: https://pixabay.com/vectors/red-science-diagram-cell-biology-41576/

- How these solutions affect our cells:
  - If our blood stream is **hypertonic** relative to our cells, water moves out to dilute the blood stream
    - cells shrivel & can die



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Hypertonic dehydration occurs when an individual excretes too much water without also excreting electrolytes, leaving the fluid that surrounds cells (i.e., extracellular fluid) with a high sodium concentration.

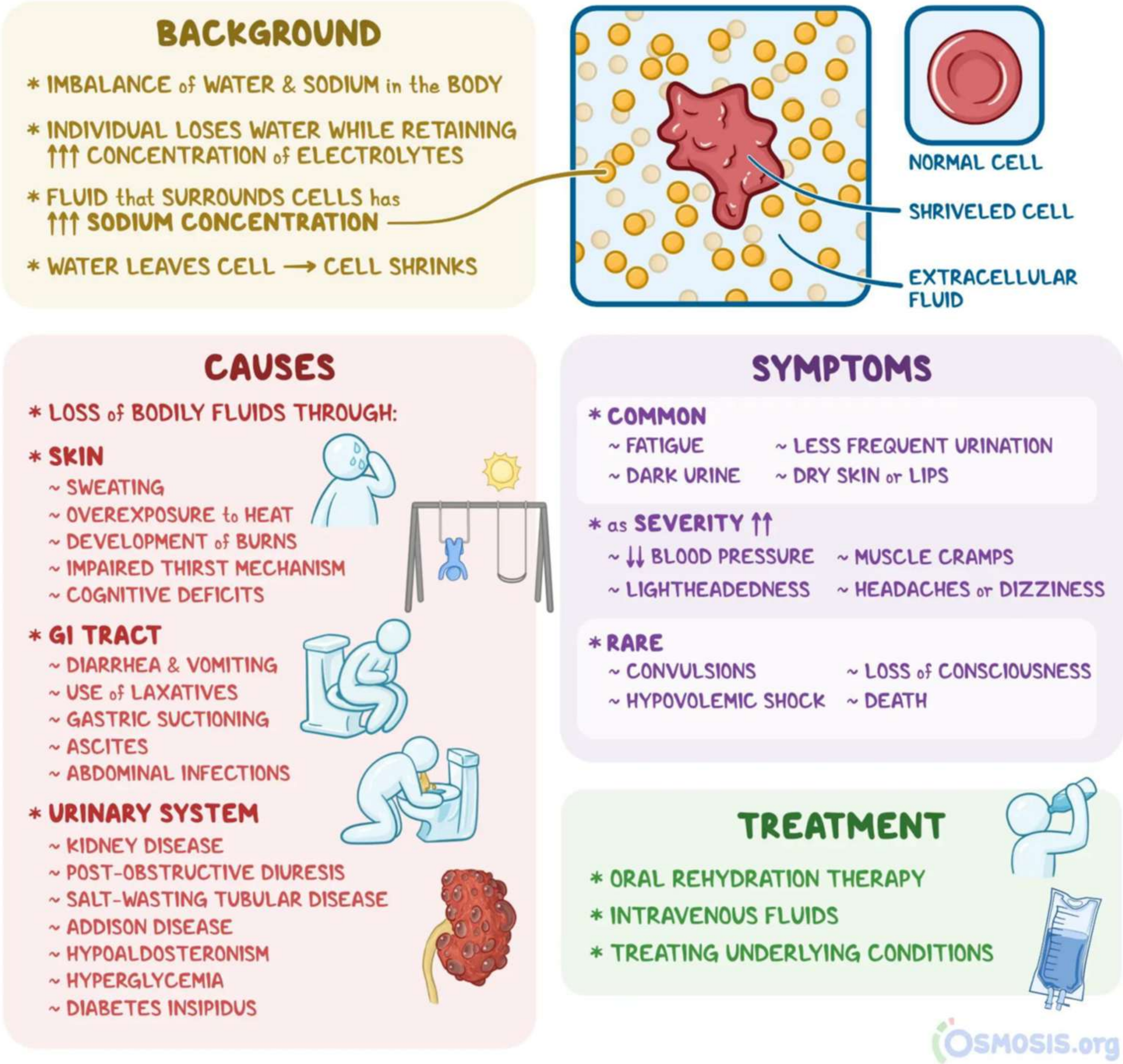


Figure: https://pixabay.com/vectors/red-science-diagram-cell-biology-41576/

- <u>How these solutions affect our cells:</u>
  - – If our blood stream is **hypotonic** relative to our cells, water moves in to dilute our cells
    - cells swell & can even burst (die)

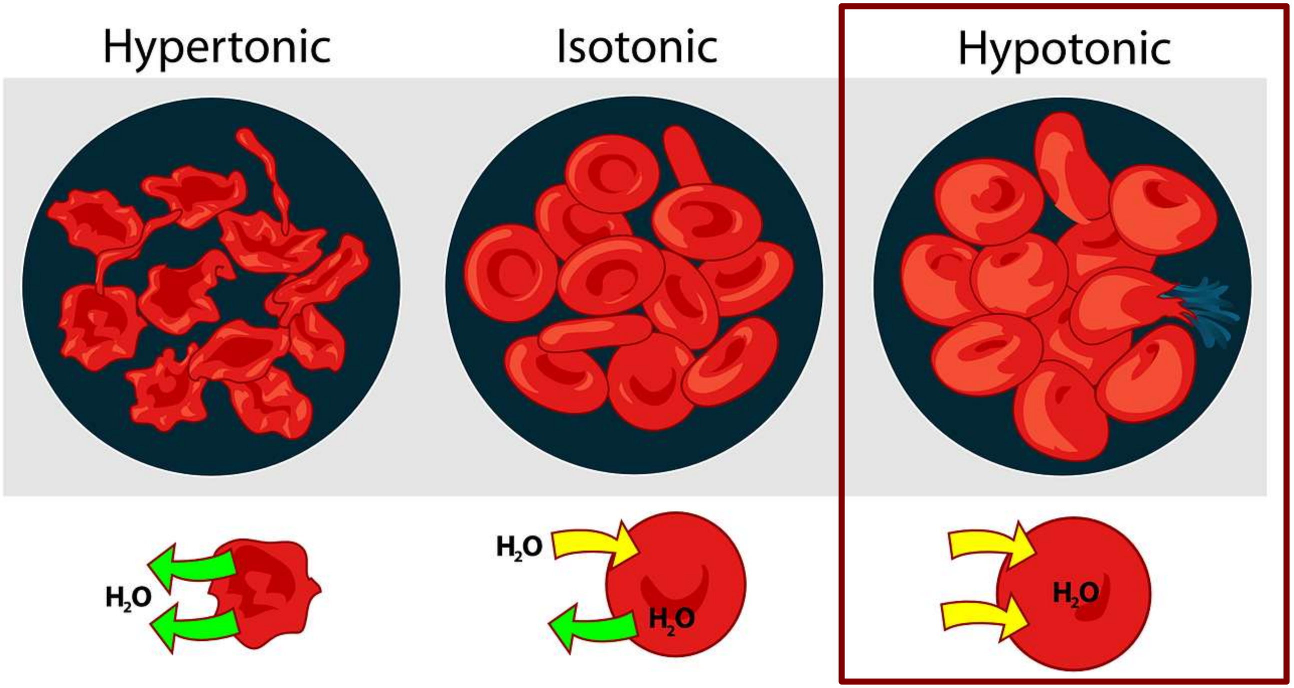


Figure: https://pixabay.com/vectors/red-science-diagram-cell-biology-41576/

Measure Serum Osmolality

Normal (280-290 mOsm/kg)

Pseudohyponatremia
- Hyperlipidemia
- Hyperproteinemia

Low (<280 mOsm/kg)

Asses Volume Status

Elevated (>290 mOsm/kg)

Hypertonic hyponatremia
- Hypeyglycemia
- Hypertonic infusions (mannitol, glycerie)

Hypervolumic
- CHF
- Cirrhosis
- Nephrotic syndrome

Hypovolumic

Euvolumic
- SIADH(Urine Osm >100)
- Psycogenic polydipsia (Urine Osm <100)
- Hypothyroidism
- Cortisol deficiency

Check Urinary Na^+

<15 mEq/L(Extrarenal loss)
- Vomitting, Diarrhea, fistulas
- Burns, sweating
- Pancreatitis, Peritonitis

>15 mEq/L(Renal loss)
- Recent diuretics
- Adrenal insuffiency
- Cerebral Salt wasting

- <u>Osmosis regulates plant cells too:</u>
  – The inside of plant cells usually have more solutes than its surroundings, which are hypotonic
    - So water enters by osmosis, generating **turgor pressure**

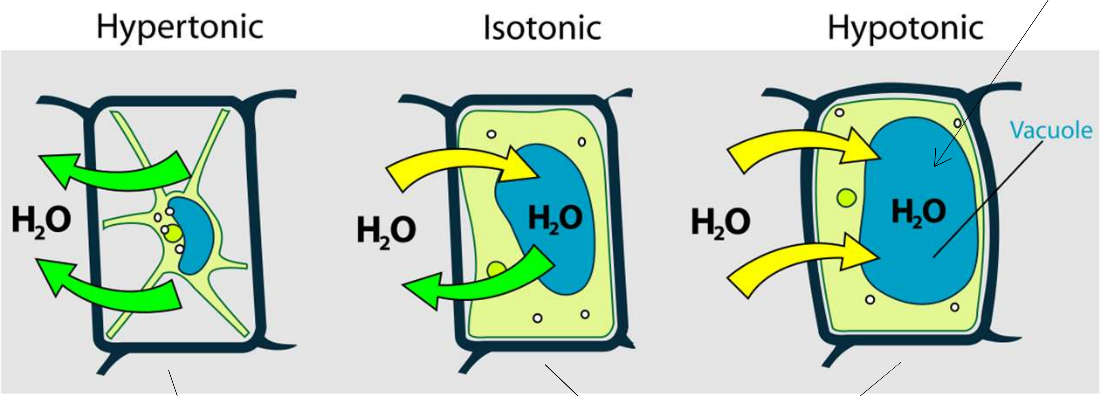


Figure: https://commons.wikimedia.org/wiki/File:Turgor_pressure_on_plant_cells_diagram.svg
Figure: https://botit.botany.wisc.edu/botany_130/anatomy/col1.html

Osmosis regulates plant cells too:

- In a hypertonic environment (e.g. if you forget to water it), water leaves the cells instead

turgor pressure in the central vacuole

- turgor pressure is no longer supporting the cells – so the plant wilts

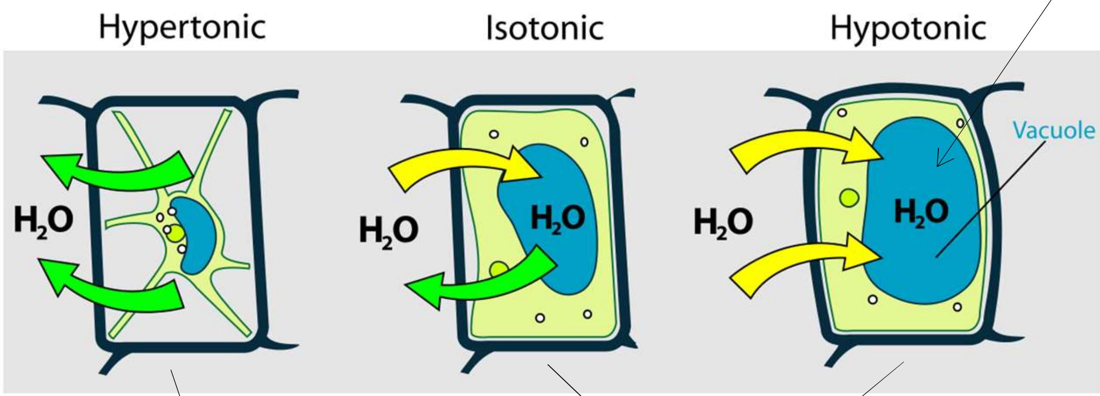


Figure: https://commons.wikimedia.org/wiki/File:Turgor_pressure_on_plant_cells_diagram.svg
Figure: https://botit.botany.wisc.edu/botany_130/anatomy/col1.html

- Osmosis vs. diffusion: which one will happen?
  - 1st we need to know if solutes can cross
    - If so, diffusion will happen
    - If NOT, osmosis will happen
  - What if both can cross?
    - Diffusion will happen until equilibrium is reached
      - Water no longer has to do any work = no osmosis

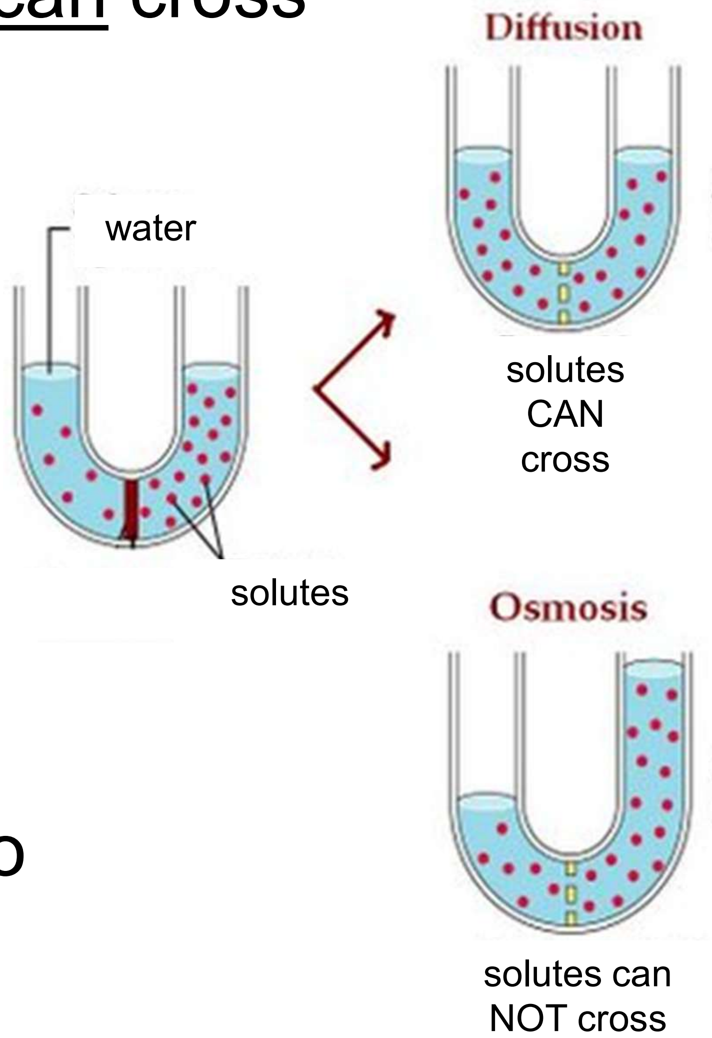


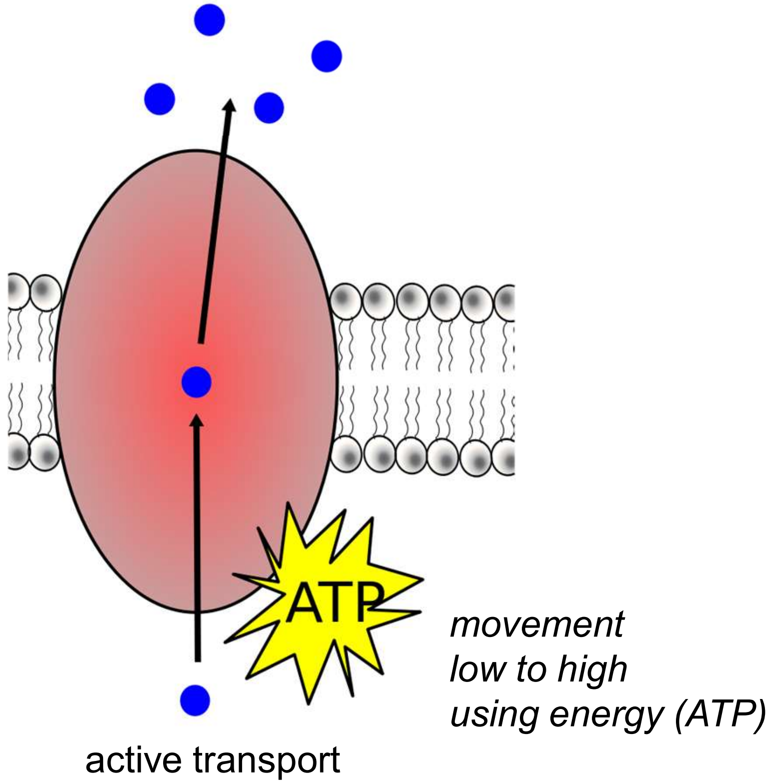
Figure: https://schsbiologyeocreview.weebly.com/picturesvideo.html

- **Energy-requiring transport**

  - Sometimes a cell needs to move molecules against their concentration gradient
    - *e.g. too much calcium can poison a cell, so it actively pumps calcium out, even if it is at a higher concentration outside the cell*

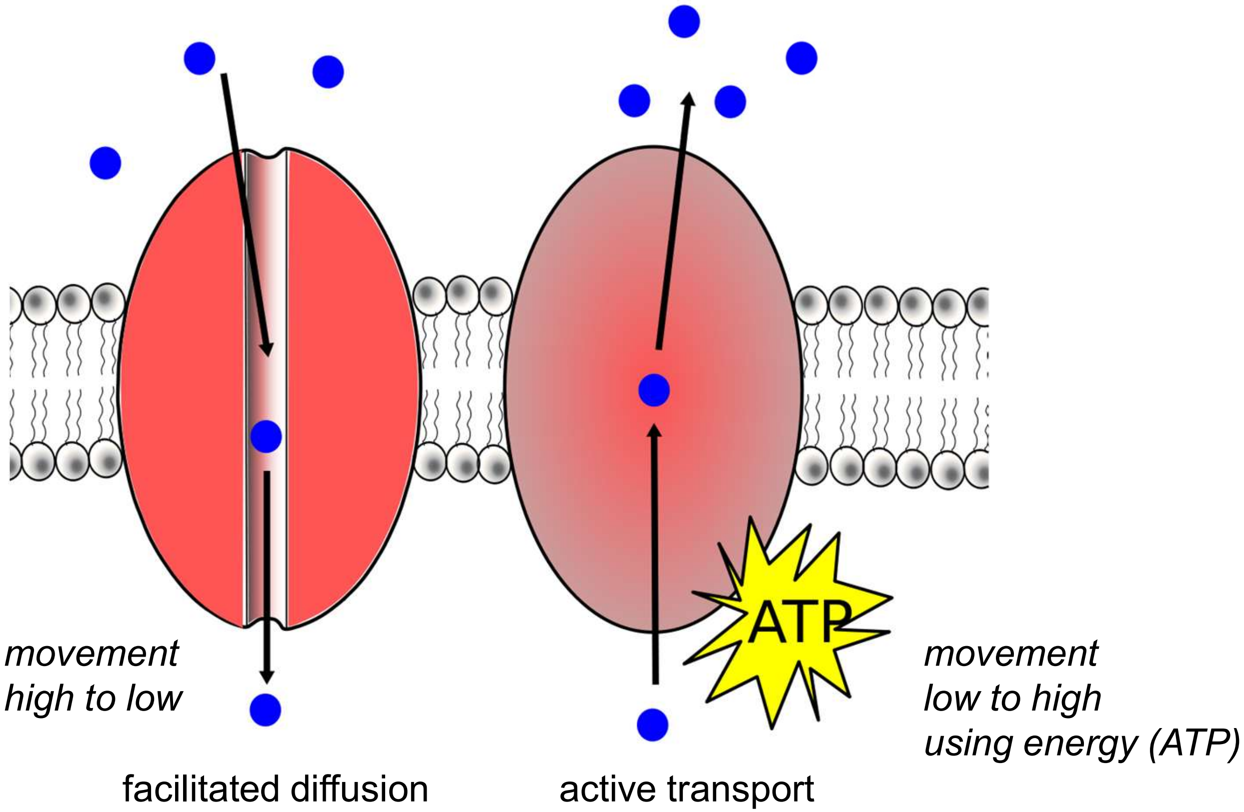
  - Sometimes a cell needs to move molecules too big to fit through proteins
    - *e.g. white blood cells of the immune system take in large bacterial cells to kill them*

- **Energy-requiring transport**:
  - cells use energy for these 3 types of transport that either move molecules against their concentration gradient or that move large molecules
    - Active transport
    - Endocytosis
      - pinocytosis
      - phagocytosis
      - receptor-mediated
    - Exocytosis

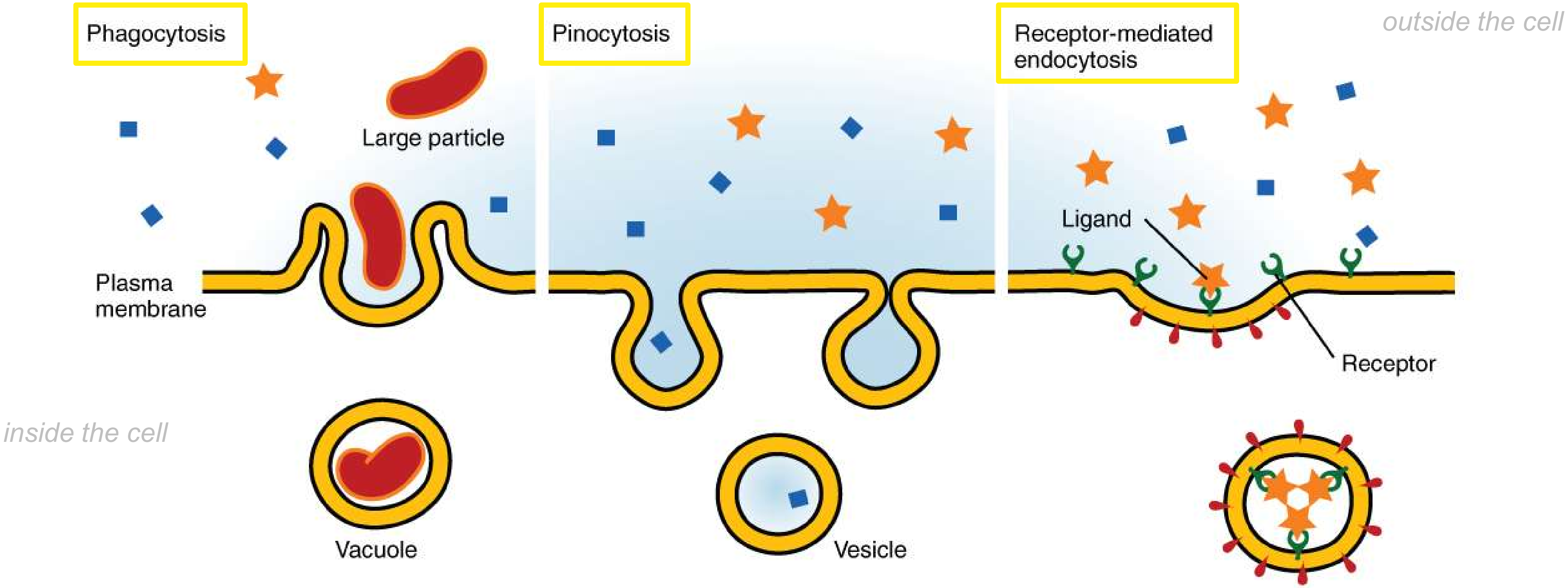


- **Active transport**
  – cells use a protein & energy (in ATP) to move molecules across cell membranes <u>against</u> their concentration gradients

too much calcium can poison a cell, so it actively pumps calcium out, even if it is at a higher concentration outside



- **Endocytosis** = into the cell
  - cells take in large molecules, food particles, or large amounts of fluid by forming a vesicle



- *3 types of endocytosis: pinocytosis, phagocytosis, & receptor-mediated*

Figure: https://commons.wikimedia.org/wiki/File:0309_Three_Forms_of_Endocytosis.jpg

- Endocytosis 1. **Pinocytosis**: moves fluids into the cell – “cell drinking”
- Endocytosis 2. **Receptor-mediated**: moves specific molecules into the cell using receptor proteins

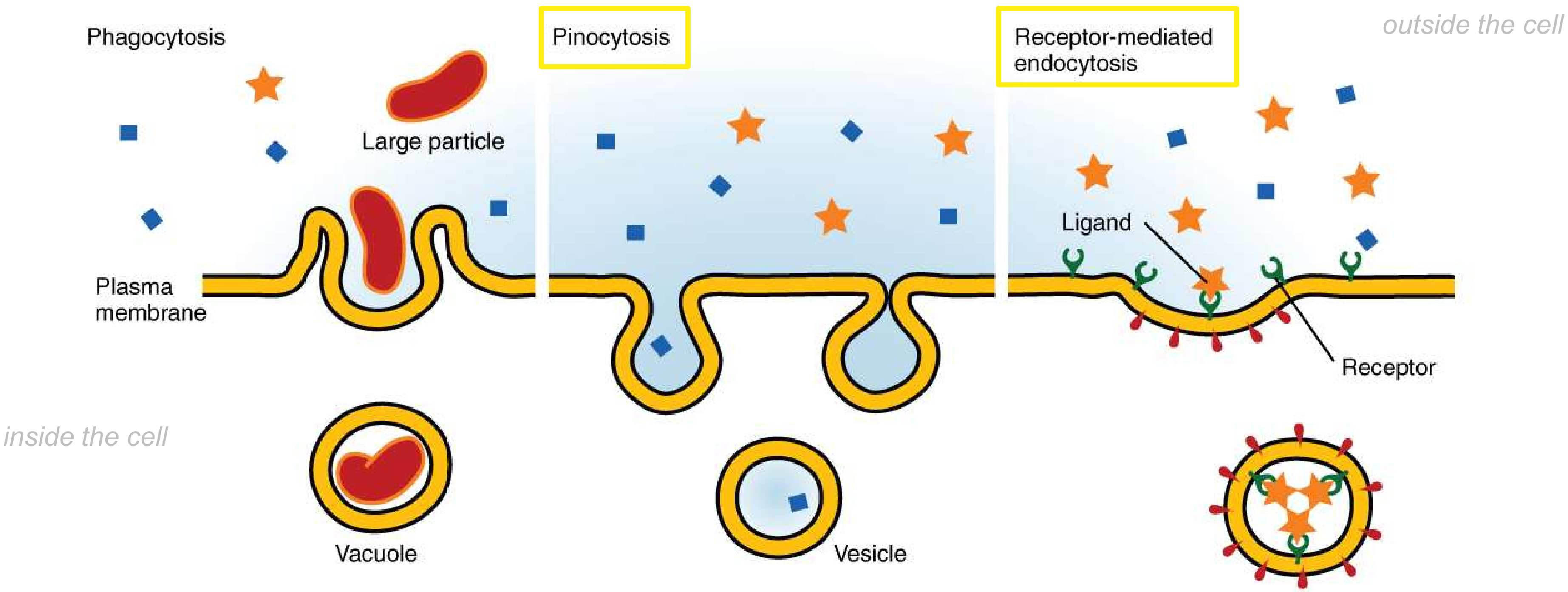


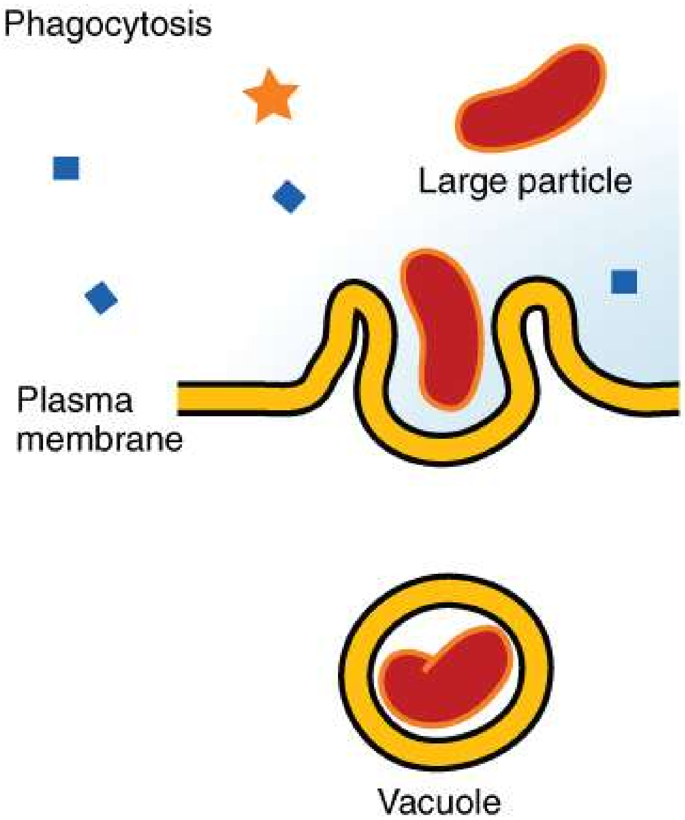
Figure: https://commons.wikimedia.org/wiki/File:0309_Three_Forms_of_Endocytosis.jpg

- Endocytosis 3. **Phagocytosis**: moves large particles into the cell – “cell eating”

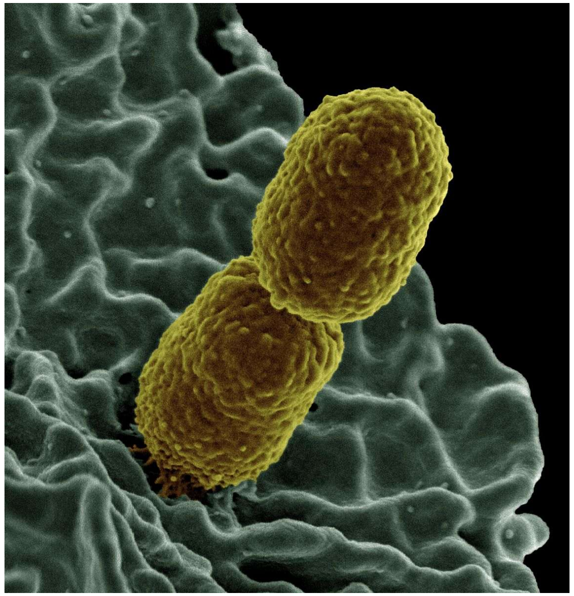
outside the cell

The cell membrane forms a pocket-like vesicle around a large particle.

The particle is transported into the cell in a vesicle.



inside the cell



A white blood cell taking in bacteria through phagocytosis

Figure: https://commons.wikimedia.org/wiki/File:0309_Three_Forms_of_Endocytosis.jpg

Figure: https://commons.wikimedia.org/wiki/File:Klebsiella_pneumoniae_Bacterium_(13383411493).jpg

- **Exocytosis** = out of the cell
  - – cell uses energy to dispose of waste or secrete substances
    - vesicles containing the material move to & fuse with the cell membrane, then allow contents to empty out of the cell

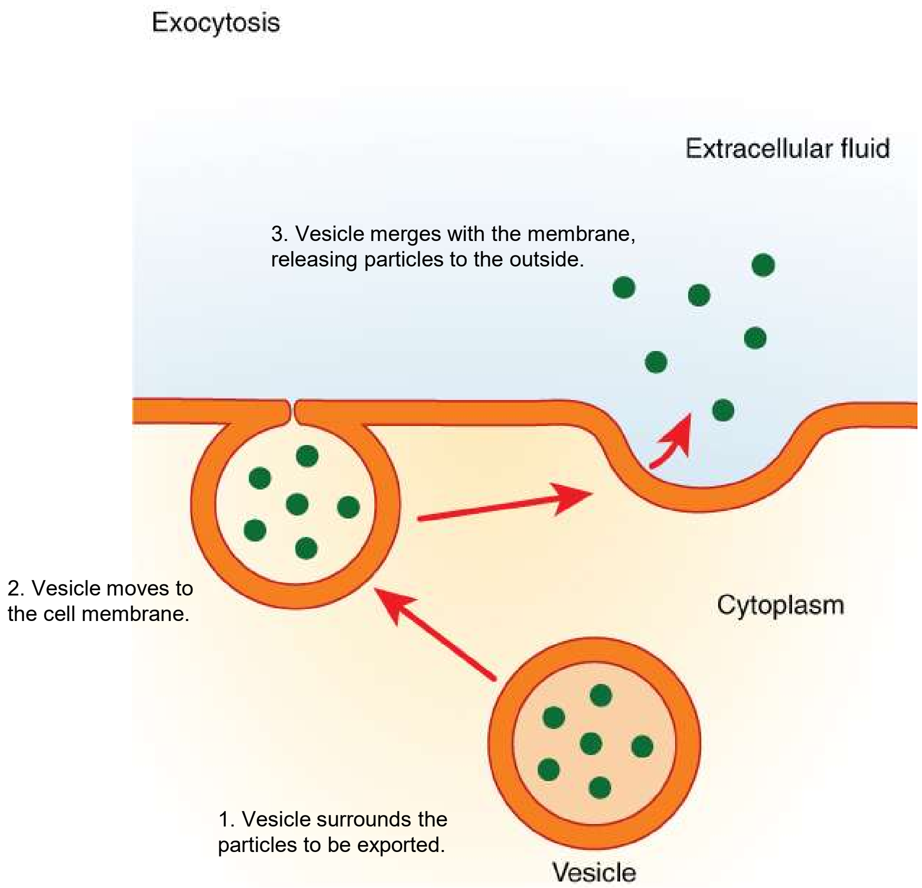


Figure: https://commons.wikimedia.org/wiki/File:0310_Exocytosis.jpg

- So then where do cells get the energy for energy-requiring transport…?

  – Coming up:

    • What is energy?

    • How do we capture energy on Earth?

    • How do we get energy out of food molecules?